

**DEVELOPMENT OF A WEB-BASED COCOA DISEASE
CLASSIFICATION MODEL WITH LEAF IMAGES USING
CONVOLUTIONAL NEURAL NETWORK**

BY

**ADEYEMI DAVID TOLUWANIMI
(20CJ027411)**

JUNE 2025

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**A PROJECT REPORT SUBMITTED TO THE DEPARTMENT OF
ELECTRICAL & INFORMATION ENGINEERING, IN PARTIAL
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COMPUTER ENGINEERING, COVENANT UNIVERSITY, OTA,
OGUN STATE, NIGERIA.**

DR. OKOKPUJIE O. KENNEDY

JUNE 2025

DECLARATION

I hereby declare that I carried out the work reported in this project in the Department of Electrical and Information Engineering, Covenant University, under the supervision of Dr. Okokpujie O. Kennedy.. I also solemnly declare that to the best of my knowledge, no part of this report has been submitted here or elsewhere in a previous application for the award of a degree. All sources of knowledge used have been duly acknowledged.

.....

(ADEYEMI, David Toluwanimi)

20CJ027411

CERTIFICATION

This is to certify that the project titled "Development of a Web-Based Cocoa Disease Classification Model with Leaf Images Using Convolutional Neural Network" by Adeyemi David Toluwanimi, meets the requirements and regulations governing the award of the Bachelor of Engineering, Computer Engineering degree of Covenant University and is approved for its contribution to knowledge and literary presentation

Supervisor: Sign: _____

Name: Dr. Okokpujie O. Kennedy Date:

Internal Examiner: Sign: _____

Name: _____ Date:

Head of Department: Sign: _____

Name: Dr. Isaac A. Samuel Date:

DEDICATION

This report is dedicated first and foremost to the Almighty God. I also dedicate this report to my parents, Engr., Mr. Adewale and Mrs. Caroline Adeyemi, who supported me during the project. To God alone be all the glory.

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TABLE OF CONTENTS

DECLARATION	i
CERTIFICATION	ii
DEDICATION	iii
ACKNOWLEDGEMENTS	iv
TABLE OF CONTENTS	v
LIST OF FIGURES	viii
LIST OF TABLES	xi
LIST OF ABRREVIATIONS	xii
ABSTRACT	xiii
CHAPTER ONE: INTRODUCTION	1
1.1 Background of the Study	1
1.2 Motivation for the Study	2
1.3 Problem Statement	2
1.4 Aim and Objectives	3
1.5 Methodology	3
1.6 Scope and Limitations of the Study	4
1.7 Project Report Organization	5
CHAPTER TWO: LITERATURE REVIEW	6
2.1 Introduction	6
2.2 Cocoa Production and Varieties	6
2.3 Cocoa Diseases	9
2.3.1 Cocoa Swollen Shoot Virus Disease	9
2.3.2 Anthracnose	11
2.4 Traditional Methods of Disease Detection	12
2.5 Artificial Intelligence in Disease Detection	14
2.5.1 Artificial Intelligence and Transfer Learning	14
2.5.2 Convolutional Neural Networks (CNNs)	15
2.6 Related Works	17
2.7 Web Deployment of Deep Learning Models	26

2.8	Chapter Summary	27
CHAPTER THREE: METHOD ANALYSIS, DESIGN AND IMPLEMENTATION		29
3.1	Introduction	29
3.2	Conceptual Framework	30
3.3	Dataset Acquisition	32
3.4	Data Preprocessing	33
3.5	Model Architecture and Selection	34
3.5.1	DenseNet-121	34
3.5.2	EfficientNet-B0	36
3.5.3	ResNet-50	38
3.6	Training Configuration	39
3.7	Evaluation Metrics	41
3.7.1	Accuracy	41
3.7.2	Precision	41
3.7.3	Recall	42
3.7.4	F1-Score	42
3.7.5	Confusion Matrix	42
3.8	Deployment Framework	42
3.8.1	Backend Implementation in Kaggle + Flask + ngrok	43
3.8.2	Frontend Interface and Deployment on Vercel	43
3.9	Experimental Setup	43
3.9.1	Hardware Environment	44
3.9.2	Software Stack	44
3.9.3	Workflow Summary	45
3.10	Chapter Summary	46
CHAPTER FOUR: RESULTS AND DISCUSSION		48
4.1	Introduction	48
4.2	DenseNet-121 Model Results	48
4.2.1	Training of DenseNet-121	48
4.2.2	Evaluation using 5-Fold Cross Validation on DenseNet-121	50

4.3	EfficientNet-B0 Model Results	56
4.3.1	Training of EfficientNet-B0	56
4.3.2	Evaluation of EffiecientNetB0 Using 5-Fold Cross Validation .	58
4.4	ResNet-50 Model	65
4.4.1	Training of ResNet-50	65
4.4.2	Evaluation of ResNet-50 Using 5-Fold Cross Validation	66
4.5	Overall Comparative Analysis of Trained Models	75
4.6	Web Application Deployment	75
4.7	Evaluation of Model Performance on Web Application	76
4.8	Chapter Summary	77
CHAPTER FIVE: SUMMARY, RECOMMENDATIONS AND CONCLUSIONS.....		78
5.1	Summary of Findings	78
5.2	Recommendations	78
5.3	Conclusions	79
REFERENCES		80

LIST OF FIGURES

Figure 3.1: Conceptual Framework for Transfer Learning and Web Deployment	31
Figure 3.2: Conceptual Framework of Cross-Validation Evaluation	32
Figure 3.3: Images of (a) healthy, (b) CSSVD-infected, and (c) anthracnose-infected leaves.	33
Figure 3.4: Diagram showing the Architecture of the DenseNet Model	35
Figure 3. 5: Comparison of different scaling methods vs. compound scaling	37
Figure 3. 6: Diagram showing the ResNet-50 Convolutional Layers	38
Figure 3. 7: Sequence diagram showing the workflow of the system	46
Figure 4.1: Confusion Matrix of Transfer Learning using DenseNet121	49
Figure 4.2: Graphs of Training Accuracy over Validation Loss and Training Loss over Validation Accuracy for Transfer Learning using DenseNet121	50
Figure 4.3: Confusion Matrix for the First Fold using DenseNet121	51
Figure 4.4: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 using DenseNet121	51
Figure 4.5: Confusion Matrix for the Second Fold using DenseNet121	52
Figure 4.6: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 and Fold 2 using DenseNet121	52
Figure 4.7: Confusion Matrix for the Third Fold using DenseNet121	53
Figure 4.8: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 to Fold 3 using DenseNet121	53
Figure 4.9: Confusion Matrix for the Fourth Fold using DenseNet121	54
Figure 4.10: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 to Fold 4 using DenseNet121	54
Figure 4.11: Confusion Matrix for the Fifth Fold using DenseNet121	55
Figure 4.12: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 to Fold 5 using DenseNet121	55
Figure 4.13: Confusion Matrix of Transfer Learning using EfficientB0	57
Figure 4.14: Graphs of Training Accuracy over Validation Loss and Training Loss over Validation Accuracy for Transfer Learning using EfficientB0	58
Figure 4.15: Confusion Matrix of Transfer Learning using EfficientNetB0	59
Figure 4.16: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 using EfficientNetB0	59

Figure 4.17: Confusion Matrix of Transfer Learning using EfficientNetB0	60
Figure 4.18: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 to Fold 2 using EfficientNetB0	60
Figure 4.19: Confusion Matrix for the Third Fold using EfficientNetB0	61
Figure 4.20: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 to Fold 3 using EfficientNetB0	61
Figure 4.21: Confusion Matrix for the Fourth Fold using EfficientNetB0	62
Figure 4.22: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 to Fold 4 using EfficientNetB0	63
Figure 4.23: Confusion Matrix for the Fourth Fold using ResNet50	64
Figure 4.24: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 to Fold 5 using EfficientNetB0	64
Figure 4.25: Confusion Matrix of Transfer Learning on ResNet50	66
Figure 4.26: Graphs of Training over Validation Loss and Accuracy of ResNet50 ...	66
Figure 4.27: Confusion Matrix for the First Fold using ResNet50	67
Figure 4.28: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 using ResNet50	68
Figure 4.29: Confusion Matrix for the Second Fold using ResNet50	69
Figure 4.30: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 to Fold 2 using ResNet50	69
Figure 4.31: Confusion Matrix for the Third Fold using ResNet50	70
Figure 4.32: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 to Fold 3 using ResNet50	71
Figure 4.33: Confusion Matrix for the Fourth Fold using ResNet50	72
Figure 4.34: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 to Fold 4 using ResNet50	72
Figure 4.35: Confusion Matrix for the Fourth Fold using ResNet50	73
Figure 4.36: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 to Fold 5 using ResNet50	74
Figure 4.37: The deployed model (a) analysing an image of an infected cocoa plant and (b) showing results.....	76

LIST OF TABLES

Table 2. 1: Table of Related Works	17
Table 3. 1: Objectives to Methods Mapping	29
Table 3. 2: Description of the dataset after pre-processing	34
Table 3. 3: The DenseNet121 Baseline Model Architecture	35
Table 3. 4: Table showing the EfficientNet baseline network architecture	37
Table 3. 5: Table showing the ResNet-50 baseline architecture	39
Table 3. 6: Table showing Sample in Training, Validation, and Testing	40
Table 3. 7: Software Tools and Technologies Used in Model Development and Deployment.	45
Table 4. 1: The Summary of the Results for DenseNet-121 Training	49
Table 4. 2: The Summary of the Results for Cross Validation on DenseNet-121	56
Table 4. 3: Summary of the Results for Transfer Learning on EfficientNetB0	57
Table 4. 4: Summary of the Results for Cross Validation on EfficientNetB0	65
Table 4. 5 : The Summary of the Results for Transfer Learning using Resnet-50	65
Table 4. 6: The Summary of the Results for Cross Validation using Resnet-50	74
Table 4. 7: Table showing an overall comparison of all trained models and evaluation	75
Table 4. 8: Results of testing the deployed model.	77

LIST OF ABRREVIATIONS

AI: Artificial Intelligence

ANN: Artificial Neural Network

CNNs: Convolutional Neural Networks

CSSV: Cocoa Swollen Shoot Virus

CSSVD: Cocoa Swollen Shoot Virus Disease

EIoU: Efficient Intersection over Union Loss

ELISA: enzyme-linked immunosorbent assay

Engr.: Engineer

FPN: Feature Pyramid Network

GDP: Gross Domestic Product

ICCO: International Cocoa Organization

ICT: Information and Communication Technology

IJACSA: International Journal of Advanced Computer Science and Applications

KNN: K Nearest Neighbours

LR: Logistic Regression

NCDC: National Cocoa Development Committee

PPR: Phytophthora Pod Rot

RF: Random Forest

SSD: Single Shot MultiBox Detector

SVM: Support Vector Machine

TEM: Transmission Electron Microscopy

ABSTRACT

Cocoa is an important cash crop of the West African farmers, and it faces a serious threat of diseases like Cocoa Swollen Shoot Virus Disease (CSSVD) and Anthracnose. The conventional techniques of disease detection are not very accurate and are sometimes time-consuming and unavailable to a smallholder farmer. This paper introduces a deep learning based cocoa disease classification system to detect cocoa diseases using leaf images based on Convolutional Neural Networks (CNNs). Three pre-trained architectures namely DenseNet121, ResNet50, and EfficientNetB0, were fine-tuned on a balanced subset of KaraAgro AI Cocoa Dataset. To analyse the models, the metrics included accuracy, precision, recall, and F1-score and it was also verified by 5-fold cross validation. The best overall performance was attained by DenseNet121 that had the highest accuracy of 90% on transfer learning, 87% on cross-validation and the most balanced metrics class-wise. Lightweight web application was formed by using a web application called Flask (backend on Kaggle), ngrok (API tunnelling), and Vercel (frontend hosting) to incorporate the selected model. The system enables the users to post or take pictures of the leaves and have their disease forecasts and agricultural prescription provided in real time. The study illustrates the real-life applicability of applying CNNs to a smart diagnosis of plant diseases and provides a sustainable and cost-efficient method of supporting low-resource cocoa production areas when required.

Keywords: Classification, Cocoa, Convolutional Neural Networks, CSSVD, Web Application.

CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Cocoa farming is the cultivation of cacao tree (*Theobroma cacao*) in order to harvest its pods containing the beans. It involves activities such as planting, nurturing, harvesting, fermenting and drying the beans before processing them. These beans are the source of the cocoa powder which is the key ingredient in chocolate production as well as other cocoa-based products.

The farming of cocoa serves as foreign exchange earner by boosting the Gross Domestic Product (GDP) in producing nations. It also serves as a means of employment and livelihood as it supports 5–6 million smallholder farmers globally, with over 1 million in Côte d'Ivoire, 800,000 in Ghana, and 300,000 in Nigeria. In 2022, cocoa exports were valued at \$8.29 billion, with Côte d'Ivoire leading at \$3.33 billion. Cocoa is also important to the chocolate confectionery industry, which recorded \$0.99 trillion in revenue in 2021 and has been projected to reach \$1.33 trillion by 2027. It also has health benefits because it contains flavanols which improve blood pressure and circulation [1].

Pests and diseases cause major economic losses in cocoa. This is particularly so in West Africa with its very large number of small-scale farms, which lack adequate pest and disease control. The overall production of cocoa has been reduced by the removal of millions trees due to virulent strains of Cocoa Swollen Shoot Virus Disease (CSSVD). Another disease, the Phytophthora Pod Rot (PPR), causes an average annual pod loss of about 40% in parts of Ghana and Côte d'Ivoire.[2] Diseases cause continuous financial loss to farmers as they face reduced income because lower yields while dealing with the increased costs that comes with managing diseases. Also, diseases impact the economy of major cocoa-producing regions like West Africa, as foreign exchange can suffer from production losses because cocoa is a key export [1],[2].

In modern times, Artificial Intelligence (AI) has rapidly transformed different aspects of daily life and various industries. The role of artificial intelligence in agriculture cannot be understated. AI, more specifically, Convolutional Neural Networks (CNNs), can be employed to detect plant diseases by analysing patterns, colours and

textures from images. The use AI-powered applications can classify plant diseases instantly, which if done right, reduces the need for human experts while making disease detection accessible to farmers. Through the identification of a specific disease, chemical use can be reduced by correctly applying pesticides and fungicides. It is also suitable for both small- and large-scale farms as models can be deployed on mobile phones and even drones.[3]

1.2 Motivation for the Study

The motivation for this research is driven by the urgent need to address the significant challenges posed by pests and diseases, particularly Cocoa Swollen Shoot Virus (CSSV), which contribute to over 40% of global cocoa crop losses. In response to these challenges, the International Cocoa Organization (ICCO), in partnership with industry leaders and research institutions, has launched a \$3.2 million initiative to tackle these issues in key cocoa-producing regions of West Africa, including Nigeria, Ghana, and Côte d'Ivoire.[4] With the involvement of the cocoa and chocolate industries, this project seeks to improve productivity by reducing losses caused by indigenous pests and diseases, such as CSSV. This research aims to contribute to these efforts by developing an AI-powered tool for early and accurate detection of cocoa diseases, helping farmers in rural areas make timely interventions and improving the sustainability of cocoa farming.

1.3 Problem Statement

Cocoa farmers face a number of challenges when it comes to identifying diseases. Many diseases do not show visible symptoms until they are progressed. Different diseases can cause similar symptoms, for example, leaf discoloration and wilting, lesions, making diagnosis difficult. Smallholder farmers lack training to identify diseases properly and rely on visual inspection without laboratory tests.

Manual methods of detecting diseases can be limiting in many ways. Farmers lack the expertise needed correctly identify diseases. Inspecting each plant manually can be very time-consuming and some infected plants can be missed. Diagnosis can be subjective, depending on human judgement, and vary from person to person.

Convolutional neural network models can outperform the traditional methods, identifying plant diseases to a very high level of accuracy. They process images almost instantly, reducing time spent on making decisions. They make an automated means of detecting diseases available to all farmers eliminate the need for professional expertise, making disease detection accessible to all farmers.

1.4 Aim and Objectives

The aim of this research is to develop a web-based cocoa disease classification model with leaf image using convolutional neural network. For the accurate detection, classification, and diagnosis of diseases in cocoa plants in order to enhance disease management practices and improve cocoa crop yield sustainability.

The objectives of this project are to:

- i. obtain a dataset of cocoa plant leaf images of different classes including the Cocoa Swollen Shoot Virus Disease;
- ii. preprocess the plant images and address any issues of inconsistent data formats;
- iii. develop models using two model optimization techniques;
- iv. evaluate the performance of the developed models; and
- v. integrate the most suitable model with the best performance into a user-friendly web application.

1.5 Methodology

In this research, will be concerned with the development of a classification model to detect cocoa diseases with leaf images using convolutional neural networks The method is based on a well-paved pipeline entailing data preprocessing, model training, performance and deployment.

The data set of this project was based on publicly released KaraAgro AI Cocoa Dataset, which included images, labelled as Healthy, Cocoa Swollen Shoot Virus Disease (CSSVD), and Anthracnose. To make learning balanced, similar number of images was chosen per each of the classes.

Transfer learning applied the fine-tuning of three pre-trained CNN models, including DenseNet121, ResNet50, and EfficientNetB0 base on the balanced dataset. The data preparation was done using standard preprocessing techniques like resizing, normalization and augmentation of images. Training was done with Adam optimizer and categorical cross entropy loss function and performance was measured by accuracy, precision, recall and f1-score. Also, 5-fold cross-validation was carried out to determine the robustness of a model.

Having considered all the models, DenseNet121 was found to be the top-performing framework. It was implemented on a real-time web application on a lightweight environment based on Flask (backend in Kaggle), ngrok (tunnelling API), and Vercel (host in the frontend). The application enables users to take a picture of cocoa leaves or upload their images to get up to date prediction as well as recommendations.

1.6 Scope and Limitations of the Study

This study aims to develop a web application to help cocoa farmers in West Africa detect and diagnose plant diseases, focusing on Cocoa Swollen Shoot Virus Disease (CSSVD) and Anthracnose. In nations like Nigeria, Ghana, Côte d'Ivoire, and Togo, these diseases endanger the viability of cocoa growing as well as the livelihoods of farmers by causing large crop losses. The objective is to develop a useful, user-friendly instrument that will help farmers detect these illnesses early so they can take action before their crops suffer significant damage.

This research aims to develop an AI-powered model using deep learning methods, particularly a Convolutional Neural Network (CNN). This model will be trained using images of healthy and infected cocoa plants with diseases like Anthracnose or CSSVD. These images will be carefully pre-processed and enhanced to ensure the model can accurately identify and categorize cocoa plants in different conditions, allowing it to work well across various environments and stages of disease.

To make this technology accessible to farmers, a mobile application will be created that integrates the AI model, enabling farmers to take pictures of their cocoa plants and receive instant feedback. Designed with ease of use, the app will be simple enough for farmers with minimal technological experience to navigate. The app will identify diseases and offer advice on how to manage them, empowering farmers with the knowledge they need to take action.

The application's effectiveness will be tested in real-world conditions through field trials with local farmers. These trials will evaluate the app's accuracy in detecting diseases, ease of use, and potential to improve crop productivity and disease management. By collaborating closely with farmers, this study aims to provide a solution that truly addresses their needs and promotes more sustainable farming practices.

Ultimately, this research seeks to give cocoa farmers the tools to manage disease better and protect their crops. By combining AI with mobile technology, the goal is to positively impact the everyday challenges farmers face, helping them improve their productivity and, hopefully, their overall quality of life.

1.7 Project Report Organization

The report is organized into five chapters, all of which fulfil a particular role in reviewing the development and evaluation of the cocoa disease classification system. In this chapter, the background of the study, problem statement, objectives, scope, significance and overview of the methodology are presented. It also shows how the report is organized. This chapter will give a review of available literature regarding cocoa cultivation, cocoa leaf diseases (CSSVD and Anthracnose), the methods of disease identification (conventional and AI-powered), convolutional neural networks, and other past research on plant disease detection and implementation. The research design, preparation of the data set, preprocessing, selection of the model, configuration training, evaluation of the model and deploying architecture of the proposed system have been described in this chapter. The chapter shows the training and validation performance of the CNN models, which architectures were compared, the outcomes of cross-validation, and the performance of the web application deployed. The conclusion chapter gives the main results of the study, concludes, summarizes the contribution to the knowledge, making practical recommendations, and reporting the limitations of the project.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

West Africa produces a big share of the world's cocoa, and its production is a significant aspect of the economy in this region. In Côte d'Ivoire, Ghana, Nigeria, and Cameroon, exporting cocoa is a keyway these countries earn profits and employ people working outside of cities. The crop helps millions of smallholder farmers make a living, supports having enough food, and plays a major role in the region's social and economic growth.

On the other hand, cocoa production and environment are seriously affected by various diseases. Because of these diseases, farmers get lower yields and better-quality beans, which affect the cocoa sector and the economies of cocoa-producing countries. It is very important to detect these diseases fast and correctly for effective control and management.

It is difficult to diagnose diseases through manual inspection by extension workers or farmers, since their results do not always match expert findings and can take a lot of time. As a result, people are now looking at advanced technologies to address these obstacles.

Lately, Convolutional Neural Networks (CNNs)—an aspect of Artificial Intelligence (AI)—have given promising results in detecting crop diseases through images. The program can detect infections from pictures, so automating the diagnosis process means it is more accurate, quick, and can scale up for use outside hospitals or clinics.

In this chapter, the main types of cocoa diseases, the usual methods for locating them, and recent improvements in applying AI for detection are covered. Past research is looked at that uses CNN and other machine learning methods for plant disease detection, focusing especially on cocoa. The aim of this review is to form the basis for the deep learning approach highlighted in the study.

2.2 Cocoa Production and Varieties

At first, cocoa grew in the Americas, but during colonial times, it was spread worldwide because Europeans wanted more chocolate. In the 1970s, Cadbury encouraged cocoa farming in India which resulted in a major increase in the land used. Yet, the sharp drop in cocoa prices in the 1980s led many farmers to stop growing cocoa which showed how unstable the world cocoa market can be [4]. Across the world, cocoa is a key product and 90% of it is used by the chocolate industry. In 2017, the total export value of cocoa beans was \$8.6 billion, while the world's chocolate market was worth \$103.28 billion and was expected to rise to \$161.56 billion by 2024 [5].

The cocoa industry in Nigeria has gone through many ups and downs. To deal with the Great Depression and other crises, the British set up marketing boards to manage cocoa prices during the colonial era. Yet, these boards usually chose to support the imperial agenda rather than the needs of farmers.[6] The oil boom in the 1970s encouraged investments in cocoa, but the industry stopped developing because of poor management and a shift to using oil money.[7] As cocoa production was decreasing, the National Cocoa Development Committee (NCDC) was formed in 1999 to enhance quality and boost production. Although some progress has been made, Nigeria's cocoa output is still below what was expected, and exports were estimated at \$800 million in 2016 [5].

Cocoa is still the main non-oil source of foreign exchange for Nigeria. The industry is growing again, as higher global prices attract both new farmers and professionals looking for better profits. Nevertheless, the sector is held back by insufficient infrastructure, restricted access to credit, and political problems. Ghana is ranked second globally in cocoa production, with an average yearly output of 800,000 metric tonnes. Cocoa makes up around 3% of the country's GDP and accounts for about 80% of what is sold in agricultural exports. It gives employment to about 6 million people, including more than 800,000 rural households [5].

The majority of cocoa produced worldwide is grown on small farms that are usually between 2 and 5 ha.[8] The typical farm in Nigeria is about 7 ha, and is often managed by older men who use little mechanization. For a lot of families, cocoa makes up the majority of their earnings, which supports their welfare, food supply, and ability to cope with hardship. [9]

Many smallholders use their hands to work, as they do not have access to sprayers and tractors. Because of this, farmers cannot always use effective methods to control pests and diseases [8]. In both Nigeria and Ghana, not all farmers can get extension services, credit, or technical help. For instance, only support from extension services, help with labour, and agrochemicals are found to boost output, but these are not usually available to small farmers.

Those who benefit from extension, credit, and inputs experience significant improvements in their productivity. In Ghana, those who belong to cooperatives are more likely to use modern farming methods. The lack of good roads, few machines, and weak institutions still hold back yields and increase poverty among small farmers [9].

The use of smartphones, ICT platforms, and deep learning in initiatives like AgroTIC and eKichabi has the potential to enhance extension services, disease monitoring, and access to markets. Nevertheless, many people still do not use digital technologies because they are not easy to get, are too expensive, or require certain skills [9].

There are four main varieties of cocoa: Forastero, Criollo, Trinitario, and Nacional. Forastero is responsible for about 80-85% of the world's cocoa production. This variety produces a lot, is strong, and has some resistance to diseases. Forastero trees can be found in various environments and produce a lot of beans, but the flavour is usually earthy and simple. Bulk cocoa is the main ingredient in commercial chocolate production, chosen for its large supply and regular taste, not for its unique flavours [10].

Criollo production accounts for less than 5% of the world's total. Criollo beans are called the "prince of cocoa" because they are aromatic and have mild floral, fruity, nutty, and little bitter notes. Since it produces very little and is easily attacked by pests and diseases, it is hard to farm on a large scale [10].

Trinitario is a mix of Criollo and Forastero, this type of cocoa was first grown in Trinidad in the 18th century. Makes up about 10-15% of the world's total production. Gives plants the ability to resist diseases and boost their yield but still offers some of the Criollo variety's fine flavours. The combination of earthy, fruity, and floral notes makes it a favourite among top chocolatiers [10].

The Nacional species originated in Ecuador and has been there for more than 3,500 years. In 1916, a disease called witches'-broom almost drove the trees to extinction, but pure-genotype trees were found again in the early 2000s; there are only a few left now. Its strong floral and fruity fragrance is appreciated, even though the yield is not high; this cocoa is classified as fine flavour. While some genetic studies group Nacional with Forastero, it is still recognized as a unique variety [10].

2.3 Cocoa Diseases

This section carries out an extensive review of the cocoa diseases addressed in this research

2.3.1 Cocoa Swollen Shoot Virus Disease

A group of badnaviruses (genus *Badnavirus*, family Caulimoviridae) native to West Africa causes CSSVD. Many species are associated with this disease, including Cacao swollen shoot Ghana M, Togo A/B, CD, Ghana N/Q, and the recently found Cacao red vein virus, and each one has its own pathogenic traits. They are small, circular, double-stranded DNA viruses (7.0–9.2 kb) that use reverse transcription to replicate and avoid being detected by the host. The virus is passed on to plants when mealybugs feed on them. Although more than 14 species are involved, the most effective ones in Ghana and Côte d'Ivoire are *Formicococcus njalensis*, *Planococcus citri*, and *Ferrisia virgata*. Mealybugs must feed for 16–24 hours to get the virus and can transfer it to other plants within a short time. Following infection, the insects can transmit the virus for as long as four days. This disease can also be spread when infected farm tools, wounds, or even rare cases of infected seedlings are used. The time it takes for symptoms to appear after infection depends on the strain and the type of cacao. It can take a long time, sometimes years, for symptoms to become noticeable, making it hard to control the disease early. When trees are infected and do not show symptoms, they may still contribute to the spread of the disease [11].

At the beginning of infection, leaves may become yellow or bleached in areas between veins, and sometimes there is red or green banding in young leaves. More symptoms on leaves are pinpoint flecks, blotches, diffused flecking, and mosaic patterns. An obvious sign is when stems, shoots, and roots swell, usually along with

deformities and cankers. Some mild viral strains can cause chlorosis without causing the leaves to swell. Trees with infections tend to be shorter than normal and have fewer and smaller pods. As the virus spreads, branches, shoot tips, and the tree itself may die. Pods could shrink, become more rounded, and become misshapen; sometimes with chlorotic or necrotic areas [12]. Sometimes, symptoms do not appear until months or even years after the infection, so it is difficult to detect the disease early. How a virus is expressed (e.g., swelling, leaf patterns) depends mainly on both the virus strain and the host plant's genes. Both leaf changes and swelling in the shoots and roots are strong signs of CSSVD and make it easier to tell apart from other diseases. Even so, some mild strains only cause problems in leaves, which is why trained observation and lab tests are needed, especially for early or symptomless cases [13].

Trees infected with CSSVD usually lose about 25% of their yield in the first year, 50% in the second, and all their yield by the third year. In most cases, infected trees die in 3–5 years, which lowers the amount of productive cocoa area. In West Africa, more than 200–300 million cocoa trees have been lost because of CSSVD [14]. The Western North region has up to 81% of its farmland affected by the disease. Ghana produced the least amount of cocoa in 20 years, mainly because of CSSVD, as well as older plantations and challenges from climate change. It takes 2–4 years for replanted areas to become productive again, so the benefits of rehabilitation are not instant [15]. CSSVD is among the world's most economically destructive viral diseases that affect plants. When cocoa trees are destroyed, farmers cannot earn a steady income, so they have less money for healthcare, schools, and new ways to support themselves. In the period from 2006 to 2010, more than 28 million cocoa trees were destroyed in Ghana to control disease. Farmers remove large areas of forest, but it takes years for trees to grow back, leaving empty space in the meantime [14].

Mealybugs feeding on the cacao plant spread CSSVD to the plant in a semi-persistent way. Among the important vectors are *Formicococcus njalensis*, *Planococcus citri*, and *Ferrisia virgata* which get infected with the virus after 20 minutes to several hours and can stay infective for days. Young mealybugs are able to move and spread the virus more easily, while adults are not; wind and ants moving in the canopy can also help them transmit the virus. CSSVD can be spread from tree

to tree by using the same pruning tools or knives, which carry infected sap. It is clear from experiments and studies that wounding plants with tools or grafting can cause infection. Suggestions have been made that CSSVD may be passed from seeds to the seedlings that grow from them. CSSV can be found in other hosts such as *Ceiba pentandra* and *Cola spp.*, but they do not transmit the virus as well. After getting the virus, mealybugs can spread it to other plants in just hours, and they may keep it for 3–4 days [12]. Mealybugs travel by crawling, riding on ants, and being carried by wind, which can move them up to 100 meters, mainly in places where trees overlap. Spread is made worse by using mechanical tools and infected seedlings, especially when trees are close and share equipment. Since mealybugs quickly spread CSSVD and infected trees can remain unnoticed, it is important to catch them early.

The best approach to control the disease is to take out all infected trees and nearby contact trees, then replace them with healthy ones [15]. Ghana's program has cleared away over 300 million trees since the 1940s, but it is still costly and takes a long time to see results, as it takes 2–4 years to replant the farms. There have been difficulties with adoption because farmers are reluctant, there isn't enough money, and policies keep changing. Even though insecticides, parasiticides, and managing shade have been tried, their success has been limited. Agroforestry with 50% shade greatly reduces CSSVD severity by changing the climate and making it harder for vectors to move. It is estimated that only a fraction of farmers use all the recommended ways to control diseases, and about half of the eradication efforts are not done correctly [16].

2.3.2 Anthracnose

The disease anthracnose in cacao is brought on by fungi from the *Colletotrichum gloeosporioides* species complex, including *C. gloeosporioides*, *C. siamense*, *C. theobromicola*, *C. tropicale*, and *C. aeshynomenes*. They first act as biotrophs by infecting living tissue and then switch to necrotrophs by killing the host and releasing spores. When it is warm, humid, and wet (temperatures between 22 and 32 °C, humidity is over 90%), spores are released, germinate, and cause infection quickly. Rain, watering, and high humidity allow the spores to travel short distances between leaves and pods [17].

Most of the time, cocoa pods have semicircular, sunken lesions that can be arranged as rings or resemble freckles. Necrotic areas frequently have acervuli, small fungal

bodies that appear as black specks or spore masses in fully developed lesions. On leaves, the first sign is a small water-soaked spot, which grows into a dark brown or black irregular patch, usually surrounded by yellow. If these spots join, they may lead to major leaf blight and tissue loss. Anthracnose can harm flowers, twigs, and petioles, leading to necrotic spots and the possible death of affected parts. Following the harvest, infected pods may rot quickly and cause hidden infections that are not seen until later. At first, the pathogen is hidden during the biotrophic stage (about 3–4 days after infection), and then it becomes visible during the necrotrophic stage (about 6–8 days), causing major damage to the tissue. Humid and rainy weather encourages the spread of lesions because moisture and spore splashing make it easier for fungus to infect the skin [18].

CSSVD and Anthracnose result in cocoa leaves turning yellow or spotted and dying in places along the stem. This intersection is often a challenge when trying to visually diagnose crops, especially for unskilled farmers and staff. Both diseases have signs of leaf yellowing and necrosis, yet CSSVD is caused by a virus and Anthracnose is caused by a fungus. Sharply defined, sunken lesions with necrotic centers and rings are typical for anthracnose, and it is often possible to see black, fungal fruiting bodies called acervuli on the affected pods or leaves. You can spot CSSVD by its swollen shoots, lack of veins, and slow growth, but Anthracnose does not have these. In CSSVD, leaves frequently turn yellow and the process is more widespread, while shoots and veins may also widen. When symptoms are similar, it can result in ineffective ways to control the disease. For example, when managing Anthracnose, you should use fungicides and cultural controls, but with CSSVD, the solution is to remove infected trees and manage insects. It is very important to distinguish between diseases early and accurately, as this helps reduce losses and allows for proper treatment, so advanced detection systems, such as molecular diagnostics and AI models, are needed. Because smallholder farmers usually do not have access to laboratory tests, they identify diseases by what they see, which can lead to mistakes. So, tools that are user-friendly and accurate are important for supporting good decisions at the farm level.

2.4 Traditional Methods of Disease Detection

Traditional methods depend on spotting signs such as leaf spots, wilting, lesions, and unusual growth. Farmers and field agronomists do this a lot because it is simple and fast. Often, diseases, nutrient deficiencies, pests, and abiotic stresses look the same, which can cause them to be misdiagnosed. Many viruses stay hidden until the disease is well advanced, so it is often hard to detect them in the early stages. The accuracy of observations is strongly affected by how well trained and experienced the observer is [19].

Light microscopy helps to see the structures of fungi, groups of bacteria, or changes caused by viruses in cells and offers evidence on the types of spores or hyphal structures in fungi. Transmission Electron Microscopy (TEM) enables the direct observation and grouping of viruses based on their shapes (rod-shaped or isometric). Quarantine programs can use TEM, but they are time-consuming and only work for viruses that are easy to tell apart by their shapes. Working on electronics requires the use of certain tools and knowledgeable staff [20].

There are also laboratory-based diagnostics which deliver precise results but require equipment and trained personnel. ELISA (enzyme-linked immunosorbent assay) involves using antibodies to spot pathogen-related proteins; it is a cheap and suitable way for testing many individuals. This is often used for virus identification, but it is less sensitive than other methods. Immunofluorescence relies on antibodies to see where pathogen proteins are located in tissues. Culturing involves growing germs in selective media; it is necessary for diagnosing fungal and bacterial infections. It is hard to control biotrophic pathogens, and plants may take a long time to show symptoms [19].

Culturing and microscopy methods take several days to weeks, so interventions cannot be made promptly. Preparing for ELISA and qPCR by collecting samples, extracting, and running the tests needs many hours or even days, so quick testing results are not possible. It is important to be skilled in spotting visual symptoms, such as spots and lesions, because this skill depends on the observer's level of expertise. When the symptoms of diseases are similar or when abiotic stress is involved, it is easy to misdiagnose a plant. Microscopy, serological assays, and PCR/qPCR are not widely used in low-resource areas since they need trained workers, special labs, and set protocols. Disease is only spotted in hosts like citrus or

cocoa after symptoms appear, which can be years after infection; early stages of the disease are not detected. Sometimes, pathogens are not evenly spread in plants, and early infections may be missed by molecular methods if the samples are not taken from the infected parts. Readings from visuals are not the same for everyone or everywhere, since they depend on light and the stage of the plant. This results in differences between raters and makes the reliability of their results low. Sometimes serological tests react with similar viruses, and PCR tests may report false results if the sample is contaminated, the design is poor, or the sample quality is not consistent [21].

2.5 Artificial Intelligence in Disease Detection

This section discusses an extensive review of the use of artificial intelligence and CNNs in detecting plant diseases.

2.5.1 Artificial Intelligence and Transfer Learning

Artificial Intelligence (AI) refers to the simulation of human intelligence processes by machines. That means that machines are able to imitate human thinking processes, mainly through machine learning and deep learning. In farming, AI analyses a lot of data, including plant pictures, records from sensors, and satellite pictures, to notice patterns and suggest possible results for the farmers.

AI can identify diseases by applying image recognition models such as CNNs, Vision Transformers, etc., to pictures of leaves or pods. These systems are accurate, usually reaching 95 to 99% and can detect objects in real time without damaging them. AI can also be combined with sensors and drones to collect information such as temperature, humidity, and soil moisture, and then forecast diseases by noticing changes in the environment. This integration helps create models that predict diseases so that intervention can happen promptly. Light models like MobileNet can work on phones or drones for on-the-spot disease identification in areas that have no internet [22],[23].

AI helps detect diseases before they are obvious to the human eye, making it possible to handle them early. AI can diagnose problems in many places consistently, lowering the risk of mistakes made by humans. In low-resource areas, mobile AI apps make it possible for farmers in Nigeria to use technology instead of relying on

labs. When AI is used on specific disease areas, it supports precision agriculture and reduces the use of extra agrochemicals [22].

Transfer learning involves taking a neural network model that has been pre-trained on a large-scale dataset and fine-tuning it to perform a specific task. A neural network that has already been learned from a big dataset like ImageNet can be modified for a specific purpose, for example, to detect plant diseases. It uses the knowledge stored in early network layers such as edge and texture detectors to improve performance, mainly when there is not much specialized data available [24].

Models trained on ImageNet learn basic features like shapes and textures that can also be found in plant disease images. This makes it possible to detect diseases accurately after retraining the classifier layers. Building deep networks usually requires starting with tens of thousands of images that have been labelled. Using transfer learning makes it possible to get accurate results with fewer samples. Beginning with a pre-trained model instead of training one from the beginning also saves a lot of time [24].

2.5.2 Convolutional Neural Networks (CNNs)

A Convolutional Neural Network (CNN) is a deep learning model designed to process grid-structured data like images. Instead of normal neural networks, CNNs use convolutional layers to discover how features are arranged in an image, making them perfect for image recognition like detecting diseased leaves.

There are four main types of layers: convolutional, activation, pooling, and fully connected layers. Convolutional layers use filters to look for edges, textures, or shapes in the image, which form the main parts of visual understanding. Activation layers introduce non-linearity, meaning networks can handle non-linear patterns and make them easier to learn. Pooling layers shrink the size of the feature map, keeping important features and making the model work more efficiently. Fully connected layers have the role of classifier, flattening features into decisions such as “healthy” or “diseased”.

CNNs learn on their own to notice things such as chlorosis instead of good colour, abnormal leaf shapes, and uneven patches or mold. These signs are connected to the symptoms of plant diseases and play a key role in making an accurate diagnosis.

CNNs automatically learn important features from the data they are trained on.. They perform well in visual tasks, even when conditions and backgrounds are different. With transfer learning, new models can be developed by applying what was learned in previous situations such as crops or types of diseases.

2.6 Related Works

This section discusses reviewed works similar to this research. Table 2.1 shows an overview of other related works.

Table 2. 1: Table of Related Works

	Author(s) & Year	Dataset Size	Model(s) Used	Testing Ratio	Validation Ratio	Testing Ratio	Performance Metrics	Used Leaf Images	Cross Validation	Web App	Mobile App
[25]	Mamadou et al., 2023	312 images	MobileNetV2, SVM (Support Vector Machine), Logistic Regression (LR), K Nearest Neighbours (KNN), Random Forest (RF), and XGBoost.	-	-	-	Accuracy: 86.04% Precision: 85.88% Recall: 86.03% F1-Score: 85.88%	No	No	No	No
[26]	Soh et al.,	4,390	Custom CNN,	70%	20%	10%	Accuracy:	No	No	No	No

	Author(s) & Year	Dataset Size	Model(s) Used	Testing Ratio	Validation Ratio	Testing Ratio	Performance Metrics	Used Leaf Images	Cross Validation	Web App	Mobile App
	2024	images	VGG-16, EfficientNetB0, ResNet50, and LeNet-5.				91.79% Precision: 91.79% Recall: 91.79% F1-Score: 82.08%				
[27]	Opoku et al., 2023	1,400 images	Support Vector Machine (SVM), Random Forest, Decision Tree, Naive Bayes, and an Artificial Neural Network (ANN)	70%	10%	20%	Accuracy: 85% Precision: 85% Recall: 85% F1-Score: 83%	No	No	No	No

	Author(s) & Year	Dataset Size	Model(s) Used	Testing Ratio	Validation Ratio	Testing Ratio	Performance Metrics	Used Leaf Images	Cross Validation	Web App	Mobile App
[28]	Basri et al., 2020	615 images	A custom deep learning model.	-	-	-	Accuracy: 84.87%	No	No	No	Yes
[29]	Rodriguez et al., 2021	-	Support Vector Machine (SVM), Random Forest, and an Artificial Neural Network (ANN)	-	-	-	Accuracy: 75%	Yes	No	No	No
[30]	Ferraris et al., 2023	312 images	YOLOv5m	85%	-	15%	Accuracy: 56.11%	No	Yes	No	No
[31]	Cagadas et al., 2024	2,000 images	VGG19	-	-	-	Accuracy: 88.75% Precision: 86%	Yes	No	No	No

	Author(s) & Year	Dataset Size	Model(s) Used	Testing Ratio	Validation Ratio	Testing Ratio	Performance Metrics	Used Leaf Images	Cross Validation	Web App	Mobile App
							Recall: 89% F1-Score: 89%				
[32]	Arakeri et al., 2024	764 images	Traditional Conv2D, ResNet50, VGG16, MobileNetV2, InceptionV3 and EfficientNetB1	80%	10%	10%	Accuracy: 97.36% Precision: 92.7% Recall: 91% F1-Score: 91.84%	No	No	No	Yes
[33]	Nyarko et al., 2024	2,173 images	YOLOv5m model, enhanced by incorporating a Swin Transformer, K-means++	70%	20%	10%	Accuracy: 91% Precision: 96% Recall: 94%	No	No	No	No

	Author(s) & Year	Dataset Size	Model(s) Used	Testing Ratio	Validation Ratio	Testing Ratio	Performance Metrics	Used Leaf Images	Cross Validation	Web App	Mobile App
			algorithm, and Efficient Intersection over Union Loss (EIoU)								
[34]	M. A. Atianashie	-	A customized Convolutional Neural Network (CNN) model.	80%	20%	-	Accuracy: 86.2%	No	No	No	No
[35]	Kouassi et al., 2024	6,000 images	VGG19, DenseNet201, Inception V3, and a custom CNN, evaluated using the XGBOOST classifier.	-	-	-	Accuracy: 96.30% Precision: 91.44% Recall: 100% F1-Score: 95.53%	Yes	No	No	No

	Author(s) & Year	Dataset Size	Model(s) Used	Testing Ratio	Validation Ratio	Testing Ratio	Performance Metrics	Used Leaf Images	Cross Validation	Web App	Mobile App
[36]	Coulibaly et al., 2020	10,000 images	-	-	-	-	Precision: 74% Recall: 84%	No	No	No	No
[37]	Kumi et al., 2022	-	SSD MobileNet V2, EfficientDet D0, CenterNet ResNet50 V2, and SSD ResNet50 V1 FPN.	-	-	-	Accuracy, Precision and Recall not reported	No	No	No	Yes
[38]	Vera, et al., 2024	567 images	AlexNet, GoogLeNet, ResNet18, MobileNetV3-Small, YOLOv3, EfficientDet-	80%	-	20%	Accuracy: 58.80% Precision: 60% Recall: 45%	No	No	No	Yes

	Author(s) & Year	Dataset Size	Model(s) Used	Testing Ratio	Validation Ratio	Testing Ratio	Performance Metrics	Used Leaf Images	Cross Validation	Web App	Mobile App
			Lite4, and EfficientNet- Lite4.								
[39]	Lopes, et al., 2022	1,239 images	ResNet18, ResNet50, Random Forest and Support Vector Machine.	90%	6.7%	3.3%	Accuracy: 96.82% Precision: 96.85% Recall: 97.09%	No	No	No	No
[40]	Rola et al., 2024	2,000 images	Naïve Bayes, Random Forest, Hoeffding Tree, Multilayer Neural Network, and Convolutional Neural Network	80%	-	20%	Accuracy: 99%	No	No	No	No

	Author(s) & Year	Dataset Size	Model(s) Used	Testing Ratio	Validation Ratio	Testing Ratio	Performance Metrics	Used Leaf Images	Cross Validation	Web App	Mobile App
			(CNN).								
[41]	Ayikpa et al., 2024		Logistic Regression, Support Vector Machines (SVM), Random Forest, XGBoost, and AdaBoost	80%	-	20%	Accuracy: 82.60% Precision: 83.76% Recall: 82.60% F1-Score: 82.33%	No	No	No	No
[42]	Harvyanti et al., 2023	1,200 images	AlexNet, SqueezeNet, DarkNet-19, and Modified CNN	70%	-	30%	Accuracy: 98.61% Precision: 98.60% and Recall: 98.62%	Yes	No	No	No
[43]	Lomotey et al., 2023	2,828 images	SSD MobileNet V2, EfficientDet	80%	-	20%	Mean Average Precision:	No	No	No	Yes

	Author(s) & Year	Dataset Size	Model(s) Used	Testing Ratio	Validation Ratio	Testing Ratio	Performance Metrics	Used Leaf Images	Cross Validation	Web App	Mobile App
			D0, SSD ResNet50 V1 FPN, and CentreNet ResNet50 V2				83.6%				
[44]	Atuhurra et al., 2024	17,703 images	VGG16, ResNet50, and Vision Transformer (ViT)	80%	10%	10%	Accuracy: 80% Precision: 86.05% Recall: 71.01% F1-Score: 77.81%	No	No	No	No

The Table 2.1 above provides a comparative overview of existing studies on plant diseases detection employing different deep learning techniques. It highlights the different models used, dataset sizes, performance metrics reported in accuracy, precision, recall and f1-score as reported. It also identifies which studies used leaf images for predictions and which deployed the developed model onto either web or mobile environment.

One of the constants in most of the studies is lack of use of deployed systems, especially as web or mobile apps. Among the 20 studies analysed, there were less than 5 studies that tried to do a mobile deployment and not one study had a web application. Even with high value models, any attempt to retrieve their benefits is inaccessible to the end-users as farmers and agricultural extension officers unless the models are practically deployed.

Although there have been numerous studies that concern the broad aspects of plant disease detection, very few studies use cocoa-specific leaf images. In some works, the images is of cocoa pods, not leaves which have a different structure. This restricts their usefulness and applicability in cocoa disease diagnosis where this study is to be focussed on.

Even though machine learning approaches are widely applied, the cross-validation, as one of the primary techniques of promoting the robustness of models, is almost disregarded. Performance is reported on a single hold-out test set in most of the reviewed works, and this may not measure the actual generalization capacity. The dataset only has one study which reports that it uses cross-validation as an evident condition. Such a discrepancy creates doubts regarding the chance of overfitting and unreliability of the unknown data.

2.7 Web Deployment of Deep Learning Models

Deep learning-based image classification systems, both on general purpose and in the case of agriculture, have strong practical use, but only as long as they are accurate, and more importantly accessible and user-friendly by the end users. The deployment of a web process integrates a trained model into an interactive web-based application that will allow real-time prediction without having users install or learn to use any of the machine learning tools or libraries.

The initial process of web deployment is a frontend interface where user can post any image or scan such an image and a backend server to receive these images, perform the model inference and send the predictions. This architecture acts as an interconnection between complex machine learning models and convenient digital applications.

Recent research has noted the significance of web and mobile implementation of diagnosis in agriculture. Picon et al. deployed a convolutional neural network on a mobile device to classify crop diseases, and presented the consequences of deploying neural networks to enable fast and user-friendly crops disease detection in a low-able connectivity facility [45]. Kumi et al. have created an application that can detect diseases in cocoa pods on the phone in real-time using object detection models. This study proved that real time diagnosis as well as offline field supports is possible [37].

2.8 Chapter Summary

The chapter has described available literature regarding cocoa production, major illnesses of cocoa plant, and development of disease detection methods. The focus was on Cocoa Swollen Shoot Virus Disease (CSSVD) that is one of the most economically destructive diseases in cocoa cultivation and the disease called Anthracnose. Conventional methods of detection which include visual examination and manual diagnosis were determined to be time consuming, subjective, and prone to error particularly when visual replicates of disease exist. Such constraints have led to the development of data-driven solutions, especially the ones that use artificial intelligence.

Recent findings showed that Convolutional Neural Networks (CNNs) are excellent methods of plant disease classification, with architectures like DenseNet, ResNet and even EfficientNet achieving high accuracy levels across different crops. The only gap however missing is its applicability to the real world where little has been done to bring these models to the practical level applicable by the end-users such as the farmers. This paper fills this gap since just training the CNNs to classify diseases of cocoa leaves and achieve high precision is not sufficient, in this paper, it also implements the best-performing model as a web app. This will help that the solution

will be accurate as well as it will be practically applicable in field situation where there is no expert diagnosis that can be done in the region with ease.

CHAPTER THREE

METHOD ANALYSIS, DESIGN AND IMPLEMENTATION

3.1 Introduction

This chapter outlines the research methodology used in this stud. It entails all processes used in collecting data, developing and evaluating the model. It describes the research design, dataset acquisition, preprocessing techniques, model architectures, as well as training procedures, and performance evaluation metrics. The methodology includes training three different models, EfficientNetB5, DenseNet121, ResNet50. These models were evaluated, and the best model was enhanced by performing a five-fold cross validation. The proposed model was then integrated into a web application to create a tool for classifying diseases using real-time image processing. This chapter ensures that the methods align with the study's objectives to achieve reliable results. Table 3.1 shows the methods used to achieve each objective of this research.

Table 3. 1: Objectives to Methods Mapping

Objectives	Materials	Activities
To obtain a dataset of cocoa plant leaf images of different classes including the Coca Swollen Shoot Virus Disease.	The KaraAgroAI Dataset	Acquire a dataset containing images of cocoa plant with infected and healthy classes.
To preprocess the plant images and address any issues of inconsistent data formats.	Image Processing Tools like numpy	Performed image preprocessing techniques and data augmentation to remove any inconsistencies.
To select three convolutional neural network model and train them using the pre-processed images.	Deep learning Framework, e.g. TensorFlow	Select three different CNN model architectures for model development based on efficiency and accuracy.

Objectives	Materials	Activities
		Split the pre-processed dataset into training, validation, and test sets, utilizing five-fold cross-validation.
		Train each models using the pre-processed images using optimizers while monitoring validation loss to prevent overfitting.
To evaluate the performance of the trained models and identify which model has the highest performance in each subclass category.	Accuracy, Precision, Recall	Determined each model's performance and contrast the outcomes over each other.
To integrate the most suitable model with the best performance into a user-friendly web application.	Web Development Tools: HTML/CSS & JavaScript for the frontend Flask for the backend	Design a web application interface for uploading images for classification Integrate the trained model into the web application

3.2 Conceptual Framework

Figures 3.1 and 3.2 illustrate the conceptual framework that was utilized for this project.

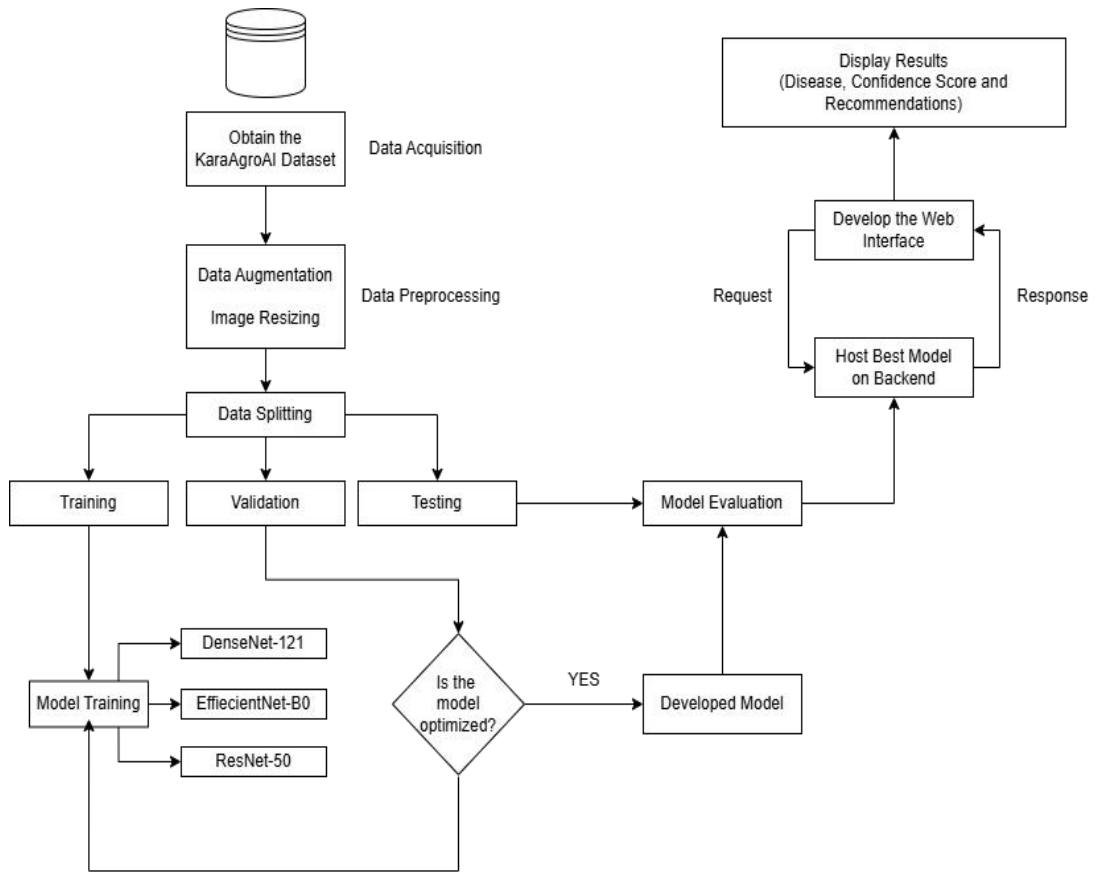


Figure 3.1: Conceptual Framework for Transfer Learning and Web Deployment

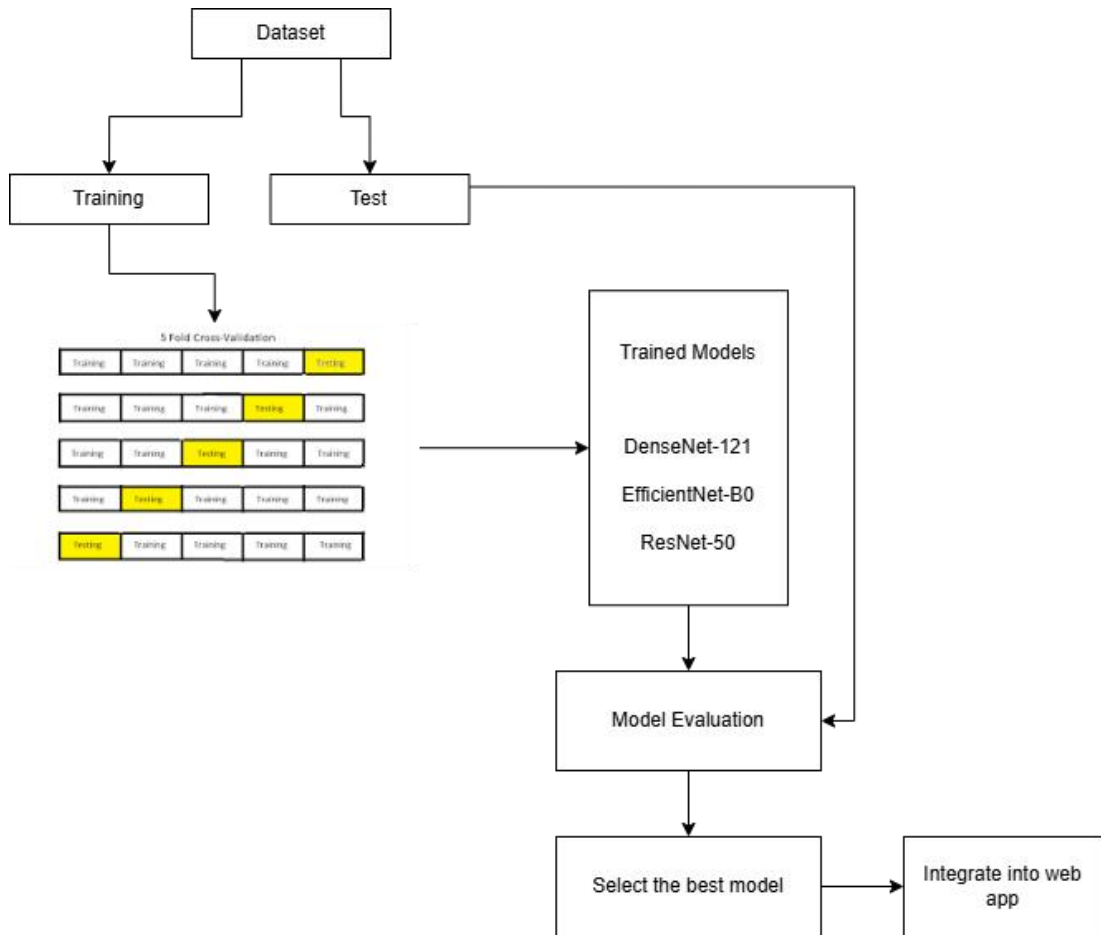


Figure 3.2: Conceptual Framework of Cross-Validation Evaluation

3.3 Dataset Acquisition

The dataset used in this research contains images of coca plant leaves, categorized into three different classes namely Healthy, CSSVD, and Anthracnose. This dataset was sourced from the KaraAgroAI Cocoa Dataset, a collection of annotated cocoa plant images, publicly available from the Havard Dataverse website. The dataset was generated to make a publicly available and accessible cocoa dataset consisting of well-labelled, adequately curated, and prepared cocoa crop imagery that would be utilized by data scientists, researchers, the general machine learning community, and social entrepreneurs in Sub-Saharan Africa and beyond to perform various machine learning experiments with the end goal of developing solutions in the diagnosis of cocoa crop diseases in-field and spatial analysis. Figure 3.2 shows image samples of (a) healthy, (b) CSSVD-infected, and (c) anthracnose-infected leaves.

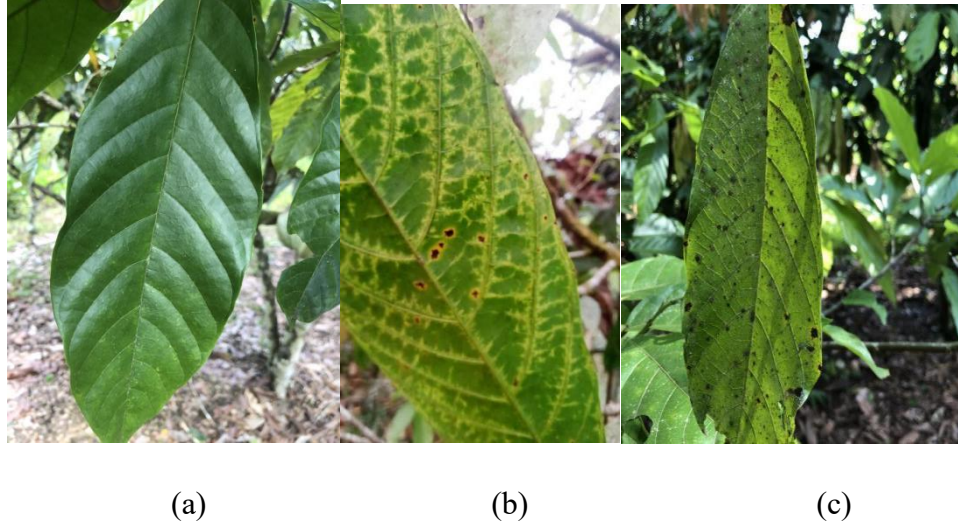


Figure 3.3: Images of (a) healthy, (b) CSSVD-infected, and (c) anthracnose-infected leaves.

3.4 Data Preprocessing

Preprocessing is the process of cleaning, transforming, and organizing data into a format that is suitable for machine learning algorithms to analyse. It is achieved through handling such problems as missing values, inconsistencies, outliers to make the data more qualitative and ready to be used in training models.

The dataset used in this study applied the following techniques:

The images used were of different sizes and had to be resized. Image resizing is the action of modifying the size of a digital image either by upscaling or downscaling intending to adjust the dimensions. All the images to 224x224 to suit all three models chosen. By down scaling the size of the images, computation is reduced to allow the training of a model on smaller images. It also made sure that there was a standardization of all the inputs to the model.

CNNs also demand a high quantity and a wide range of datasets in order to acquire the learning to be achieved to grow, this can be solved by applying methods to artificially augment the data to include more images in the dataset, moreover, to be different to one another. Data augmentation is a technique where new data samples are artificially created by applying various transformations to existing data. Such techniques include the use of rotation, flipping, manipulation of brightness and

contrast levels, zooming and cropping. Table 3.2 shows description of the dataset after pre-processing.

Table 3. 2: Description of the dataset after pre-processing

Class	Number of Images
Healthy	2584
CSSVD	2584
Anthracnose	2584

3.5 Model Architecture and Selection

This project employs three popular CNN architectures for the task of detecting and classifying the diseases in cocoa plant through leaf image analysis. These models are DenseNet121, ResNet50, and EfficientNetB0.

3.5.1 DenseNet-121

DenseNet-121 (Densely Connected Convolutional Neural Networks) is a deep convolutional neural network architecture uses dense connections to classify images.. Each layer takes in all previous layers as inputs and the layers are formed into dense blocks with transition layers in between. The result of this architecture is that an L-layer network has $L(L+1)/2$ connections. The architecture enables strong gradient flow, removes the vanishing gradient issue, reinforces the reuse of features, and reduces redundancy. It is known to be perform very high on agricultural disease detection tasks, regularly reaching the top accuracies on plant pathology benchmarks

This model architecture comprises of four dense blocks of 6, 12, 24 and 16 layer respectively. The sequences of operations implemented in each layer are BatchNorm, ReLU, 1x1 Conv, BatchNorm, ReLU and 3x3 Conv. Transition layers, placed between dense layers, consist of 1x1 convolution that perform channel compression and 2x2 average pooling for spatial down-sampling. Figure 3.3 illustrates the architecture of the DenseNet model

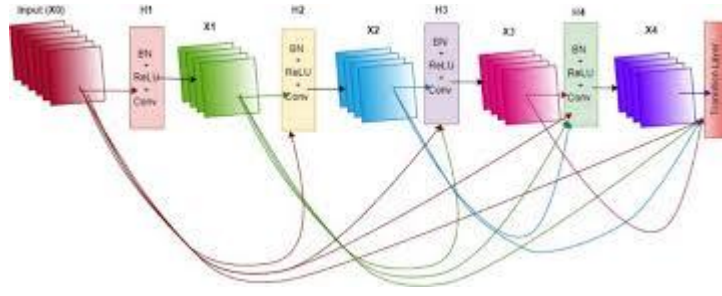


Figure 3.4: Diagram showing the Architecture of the DenseNet Model

This architecture was chosen as it improves feature reuse, which is useful when working with limited datasets. It also uses fewer parameters while maintaining a high level of accuracy, which reduces computation requirements, making it suitable for web applications. Table 3.3 explains the DenseNet121 baseline model architecture.

Table 3. 3: The DenseNet121 Baseline Model Architecture

Layers	Output Size	Description
Convolution	112 x 112	7×7 conv, stride 2
Pooling	56 x 56	3×3 max pool, stride 2
Dense Block (1)	56 x 56	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 6$
Transition Layer	56 x 56	1×1 conv
	28 x 28	2×2 average pool, stride 2
Dense Block (2)	28 x 28	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 12$
Transition Layer	28 x 28	1×1 conv
	14 x 14	2×2 average pool, stride 2
Dense Block (3)	14 x 14	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 24$
Transition Layer	14 x 14	1×1 conv
	7 x 7	2×2 average pool, stride 2

Layers	Output Size	Description
Dense Block (4)	7 x 7	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 12$
Classification Layer	1 x 1	7×7 global average pool
		1000D fully connected, softmax

3.5.2 EfficientNet-B0

EfficientNet-B0 is a convolutional neural network architecture that implements a new compound scaling approach to balance the depth, width, and input resolution of the network in a principled manner. Compared to the more typical ones, which scale arbitrarily, EfficientNet scales all dimensions with a fixed set of pre-defined coefficients. This enables it to be highly accurate and considerably low in the number of parameters and calculation costs.

EfficientNet-B0 offers blocks of mobile inverted bottleneck convolutions (MBConv), including the inductive feature of squeeze-and-excitation (SE), optimization that aids the model in dynamically paying attention to features of interest. The architecture begins with an ordinary convolution and then begins generalizing with multiple MBConv modules having different filter sizes and expansion ratios. The last layers include a 1x1 convolution, global average pooling, dropout, and a fully connected layer with softmax activation. Figure 3.4 below illustrates the difference among other scaling methods and compound scaling.

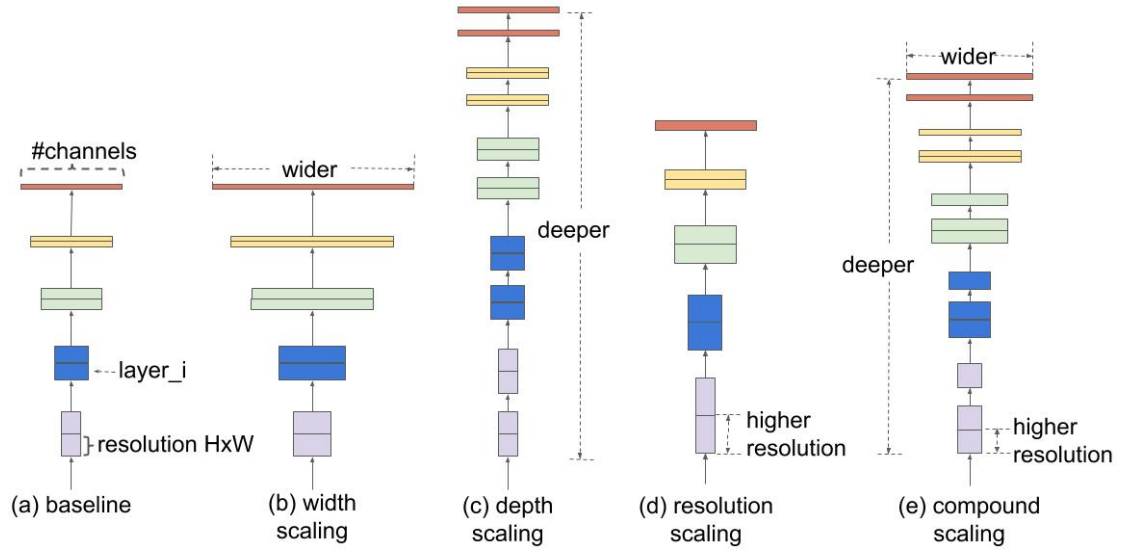


Figure 3. 5: Comparison of different scaling methods vs. compound scaling

This architecture was selected on the basis that it is efficient in terms of the trade-off between accuracy and efficiency and is perfect in a resource-limited setting like a mobile application or a web-based application. Its small size enables it to infer and use less memory, and it performs well at generalizing, which makes it adaptable to cocoa disease detection, particularly in smaller operations where a computer can be scarce. Table 3.4 explains the EfficientNetB0 baseline model architecture.

Table 3. 4: Table showing the EfficientNet baseline network architecture

Stage	Operator	Resolution	No of. Channels	No. of Layers
1	Conv3x3	224x224	32	1
2	MBConv1,k3x3	112x112	16	1
3	MBConv6,k3x3	112x112	24	2
4	MBConv6,k5x5	56x56	40	2
5	MBConv6,k3x3	28x28	80	3
6	MBConv6,k5x5	14x14	112	3
7	MBConv6,k5x5	14x14	192	4

Stage	Operator	Resolution	No of. Channels	No. of Layers
8	MBConv6,k3x3	7x7	320	1
9	Conv1x1/Pooling/FC	7x7	1280	1

3.5.3 ResNet-50

ResNet-50 (Residual Neural Network 50) is a type of deep convolutional neural network architecture and exploits residual learning to simplify the training of deep networks. Its main innovation is in the use of skip connections that propose to skip over one or more layers in the network. The shortcuts are useful in maintaining gradient flow during backpropagation thus beating the vanishing gradient problem that usually besets deeper models. Rather than learning a direct mapping, every residual block learns a residual function, the difference between the input and the desired output, and thus eases optimization and boosts the rate of convergence.

The architecture begins with a stem layer consisting of a 7 x 7 convolutional layer, which is followed by a 3 x 3 max pooling layer. It is then constituted by four convolutional blocks, which consist of bottleneck residual blocks, totalling 50 layers. A bottleneck block contains 3 convolution layers: 1x1 convolution with the goal of reducing dimensionality, 3 x 3 convolution aiming to extract spatial features, and 1x1 convolution to restore dimensionality. Each layer is followed by the Batch Normalization and ReLU activation, with a shortcut connection that adds input to the output before the final activation. The stages are in a [3, 4, 6, 3] block arrangement.

Figure 3.5 below illustrates the architecture of ResNet50 and its layers.

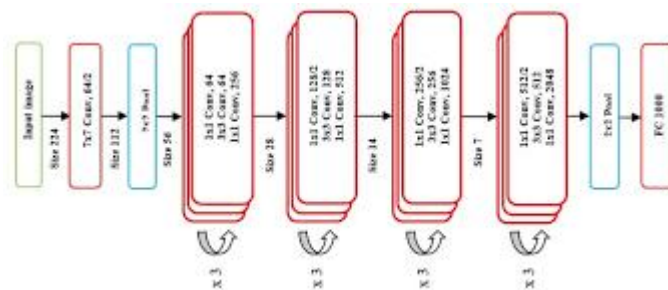


Figure 3. 6: Diagram showing the ResNet-50 Convolutional Layers

ResNet-50 was chosen as a model in this study because it scales deep networks without experiencing performance penalties and hence would work well to capture complex patterns in the symptoms of plant diseases. In terms of size, it is bigger than networks such as EfficientNet-B0 or DenseNet1-21, but still a powerful base architecture in most image classification tasks. Its compatibility with transfer learning, particularly with large-scale data sets like ImageNet, allows it to adapt successfully to moderate-scale domain-specific datasets. Also, it can run on systems with standard GPUs or high-end CPUs and is very suitable for web and mobile deployment. Table 3.5 shows the ResNet-50 baseline architecture

Table 3. 5: Table showing the ResNet-50 baseline architecture

Layer Name	Output Size	Process
Conv1	112×112	7x7,64, stride 2
Conv2_x	56×56	3x3 max pool $\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$
Conv3_x	28×28	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 4$
Conv4_x	14×14	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 6$
Conv5_x	7×7	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$
	1 x 1	average pool, 1000-d fc, softmax

3.6 Training Configuration

The training process was to optimize the chosen pre-trained CNN models of DenseNet121, ResNet50, and EfficientNetB0 options on the cocoa leaf data. This

was based on adapting such models that had initially been trained on the ImageNet dataset, to be effective in distinguishing between three classes.

Each of the models has been trained with an Adam optimizer, a common optimizer used in deep learning as it supports adaptive learning rate and comes with good convergence characteristics. The categorical cross-entropy loss was applied since it suits the categorization based on multi-class scenarios with one-hot encoded labels.

The training parameters were as follows:

a) Batch size:

This value was selected to balance computational efficiency and model stability during gradient updates.

b) Epochs: 10

The number of epochs was relatively low but adequate, as transfer learning was applied; thus, the base weights are pre-trained. Early stopping was also employed to control training, where training was stopped when validation loss failed to improve in 3 consecutive epochs.

c) Learning Rate: Adam was initialized to a 0.001 learning rate. Learning rate scheduling was utilized in some folds, and the learning rate would decrease when the validation loss stops improving.

d) Validation Split: In transfer learning, the validation set was reserved by ensuring 20% of the training set.

Table 3. 6: Table showing Sample in Training, Validation, and Testing

Set	No. of Images
Training	6201
Validation	775
Test	776

Table 3.6 above shows the number of samples used in each data split set. Besides the simple train-validation splits, the balanced dataset was used to calculate 5-fold cross-validation. The dataset was split into five equal parts; after each iteration, one portion

was considered the validation set, and the rest four were regarded as the training sets. Such an approach enabled the model to be trained and tested on various subsets of the data, making the training and testing process much more robust and limiting the level of variance related to the performance estimation.

The implementation of all training experiments was performed with the help of the TensorFlow and Keras libraries on a computer with a GPU accelerator, which will provide efficient computation.

3.7 Evaluation Metrics

In order to get a detailed analysis of the performance of the model, a number of standard classification measures were used. These measures will give information on the class's accuracy as well as the effectiveness of the model in each of the classes.

The evaluation metrics used in this study include:

3.7.1 Accuracy

Accuracy is the most basic metric and represents the percentage of correctly predicted samples of the total number of predictions. Though the accuracy proves helpful in initial evaluation, it is misleading in the case of disparity between the classes. Equation (3.1) shows how accuracy is calculated.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \quad (3.1)$$

3.7.2 Precision

Precision evaluates the proportion of correct positive predictions. For each disease class, it determines the number of images that were successfully predicted to belong to that disease class. Precision must be high, meaning that the false-positive rate is low since the misclassification of healthy plants may result in a false trigger of interventions in the case of an agricultural setting. Equation (3.2) shows the mathematical interpretation of precision.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \quad (3.2)$$

3.7.3 Recall

Recall measures the proportion of actual positive cases that were correctly identified. In this study, it shows how accurate the model correctly detects all diseased leaves, which is important for reducing overlooked cases of CSSVD and Anthracnose. Equation (3.3) shows how recall is calculated.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \quad (3.3)$$

3.7.4 F1-Score

The F1-score is the harmonic mean of precision and recall. It is a more balanced metric used in cases where there is a trade-off between false positives and false negatives, which is common in disease classification tasks. Equation (3.4) shows the formula for calculating this metric.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3.4)$$

3.7.5 Confusion Matrix

A confusion matrix was obtained with the visualization of the number of true positives, false positives, true negatives, and false negatives for each of the classes. It gives an overview of the misclassification patterns and class-wise performance that allows determining which classes the model does not distinguish well.

The evaluation metrics were calculated on both the validation set and in each of the folds in 5-fold cross-validation. This enabled a sound analysis of performance and reasonable comparison of models. Per-class and macro-averaged metrics were given to make sure that it was not a biased performance in terms of a specific class.

3.8 Deployment Framework

Following the assessment of performance by the trained models, DenseNet121 was seen to be the model that yielded best performance in terms of accuracy, precision, recall, and F1-score. The model was implemented in a web-based interface so that it can be utilized by end users, especially cocoa farmers, to enable them to use it in practical terms of uploading a cocoa leaf picture in the system and getting instant answers. The app was deployed using a hybrid architecture involving Kaggle Notebooks, ngrok tunnelling, and Vercel hosting.

3.8.1 Backend Implementation in Kaggle + Flask + ngrok

The backend was created in a Kaggle Notebook using Flask to receive uploaded images and return disease predictions. Ngrok was used to expose the public API endpoint of the local Flask server because Kaggle does not support public URLs. This allowed external access to the "/predict" endpoint by the frontend.

The trained model "DenseNet121_Cocoa_diagnosis.keras" was loaded into the memory. Online-uploaded images were resized (224×224), normalized, and reshaped to the input of the model. The model generated a prediction with a confidence value and returned the class label in a JSON object. Each prediction was mapped to a set of recommendations including HTML links to agricultural resources.

3.8.2 Frontend Interface and Deployment on Vercel

The user interface was built using HTML, CSS, and vanilla JavaScript. Then deployment was carried out on Vercel, a cloud hosting platform for static websites. From the front end, users to upload an image or take a real-time photo using their camera. A preview of the image will be displayed after upload. The image was submitted to the backend through a fetch-based POST request. The frontend receives from the backend, the predicted disease class, confidence score, and recommended action., and displays it to the user.

The app also has a dark/light theme toggle to provide users with visual comfort. It also shows a history of previous diagnosis with 10 most recent results displayed with their image, time, diagnosis, confidence and advice.

3.9 Experimental Setup

This section explains the computation platform, software libraries and tools involved in training, testing and deploying the model. The reproducibility of all experiments was secured, and all experiments were carried out with the help of free and open-source tools.

3.9.1 Hardware Environment

Model training testing and evaluation were done on a local computer where:

- a. CPU: Intel core
- b. RAM: 16 GB
- c. GPU: NVIDIA GeForce GTX 1660 Ti
- d. OS: window 10

The model deployment operated on Kaggle notebooks where a Tesla P100 graphics card with 16GB VRAM will be available mostly to host a trained model through a Flask backend that is exposed through ngrok.

3.9.2 Software Stack

This section lists the software employed tools and technologies employed including programming languages, libraries, and frameworks. Table 3.7 outlines the software tools implemented in this project.

Table 3. 7: Software Tools and Technologies Used in Model Development and Deployment.

Component	Version/Tool Used
Programming Language	Python 3.10+
Deep Learning Framework	TensorFlow 2.x, Keras API
Data Processing	NumPy, Pandas, Pillow (PIL)
Visualization	Matplotlib, Seaborn (for optional plots)
Web Backend	Flask
API Tunnelling	ngrok (to expose backend API on Kaggle)
Frontend	HTML, CSS, JavaScript
Hosting Platform	Vercel (Frontend)

3.9.3 Workflow Summary

Figure 3.6 shows a sequence diagram explaining the workflow of the proposed system

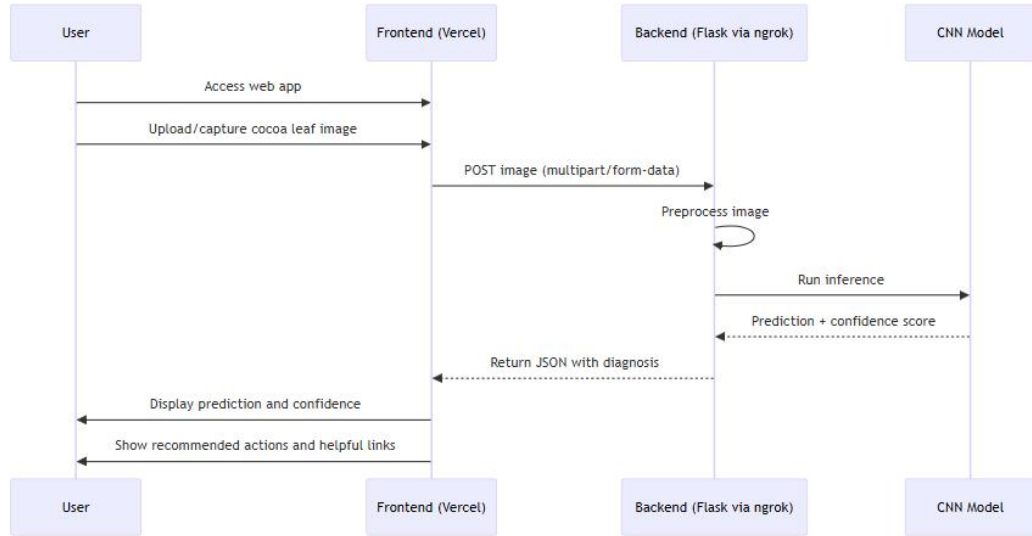


Figure 3. 7: Sequence diagram showing the workflow of the system

3.10 Chapter Summary

This chapter introduces the proposed method by giving the research plan to create a classification system of cocoa leaf disease with the help of convolutional neural networks. It provides the description of the way how a labelled dataset with healthy, CSSVD-infected leaf and anthracnose-infected cocoa leaf images were obtained using the KaraAgroAI dataset. Preprocessing tasks which included resizing of all images to be 224 x 224 pixels and data augmentation using methods of rotation, flipping, and changing brightness were carried out to enhance consistency. The decision to use three CNN architecture designs; DenseNet121, EfficientNetB0, and ResNet50, was informed by factors such as performance, accuracy, size of networks, and speed of networks. To make this robust, these models were assessed with a five-fold cross-validation after being trained with transfer-learning. In terms of metrics of accuracy, precision, recall and F1-score, DenseNet121 recorded the best performance throughout.

The model that performed most excellently is one that has been implemented into a web application that works in real-time on disease diagnosis. Backend was written in Flask and uploaded to Kaggle, and ngrok managed to allow the Flask server on the internet. The HTML, CSS and JavaScript based frontend was hosted on Vercel to enable the user to upload/capture images of cocoa leaves. After submitting an image,

it will be delivered to the backend API to be predicted, and the results of the diagnosis with confidence values, and propositions of what can be done will be shown to the user. This chapter also describes the software libraries, and hardware required to run the system, and inferred that the implementation could be reproducible.

CHAPTER FOUR

RESULTS AND DISCUSSION

4.1 Introduction

The chapter entails an in-depth discussion of the training and testing outcomes of three convolutional neural network models, including DenseNet121, EfficientNetB5, and ResNet50, in detecting cocoa disease. All models were assessed on the same data in two training strategies: transfer learning and K-fold cross-validation. Model effectiveness was measured using the performance metrics of accuracy, precision, recall, and F1-score. The obtained results are compared to other models and training approaches.

4.2 DenseNet-121 Model Results

DenseNet model is a pre-trained convolutional neural network that has been proven to be effective in processing image data especially in medical imaging because it generalizes well on small datasets and captures deep, diverse features that are useful in improving disease detection. In this project, the DenseNet121 model has been utilized in the task of cocoa disease classification with two training methods: transfer learning and K-fold cross-validation with identical dataset and parameter parameters.

4.2.1 Training of DenseNet-121

The model got the overall accuracy of 90% on 776 samples. Across all the classes, the highest recall was obtained with Healthy with a value of 0.94 meaning that the model was most sensitive in predicting healthy plant samples. Anthracnose had the highest precision of 0.95, which implied that when the model spoke of anthracnose, it was most likely to be correct. CSSVD despite being a little lower in precision and recall than the other classes, still balanced at 0.89 in both measures. Both the macro and weighted average values of precision, recall, and F1-score were noted as 0.90, indicating the balanced nature of the model in terms of classifying the data across the three classes. Table 4.1 shows the training results for this model. Figures 4.1 and 4.2 show the confusion matrix and training and validation graphs respectively.

Table 4. 1: The Summary of the Results for DenseNet-121 Training

Disease/Metric	Precision	Recall	F1-score
Anthracnose	0.95	0.88	0.91
CSSVD	0.89	0.89	0.89
Healthy	0.88	0.94	0.91
Accuracy			0.90
Macro Average	0.90	0.90	0.90
Weighted Average	0.90	0.90	0.90

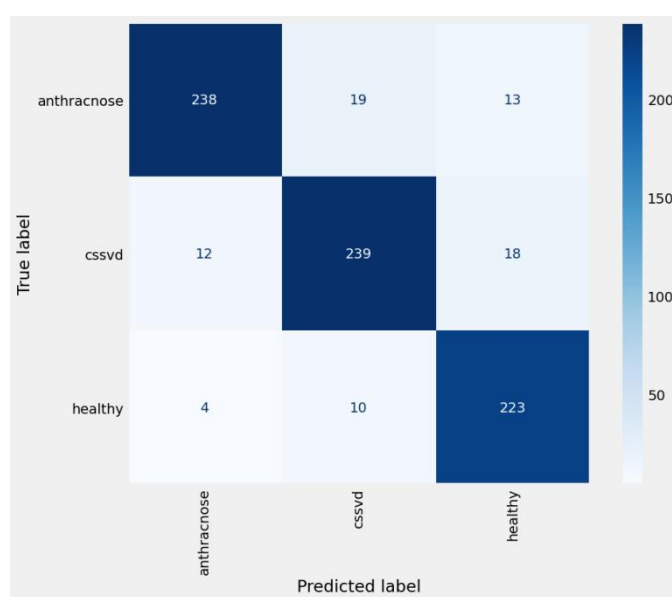


Figure 4.1: Confusion Matrix of Transfer Learning using DenseNet121

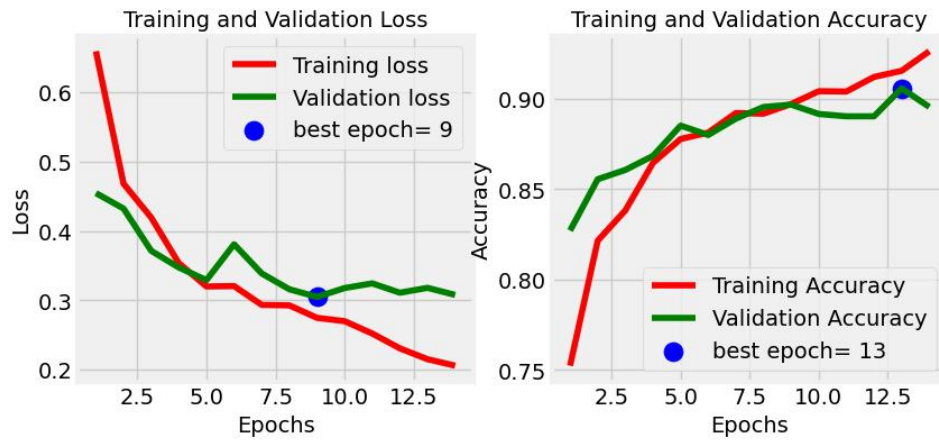


Figure 4.2: Graphs of Training Accuracy over Validation Loss and Training Loss over Validation Accuracy for Transfer Learning using DenseNet121

4.2.2 Evaluation using 5-Fold Cross Validation on DenseNet-121

This section focuses on the performance of the DenseNet121 model. The model was trained on the five-fold cross-validation method, where the data was split into five equal subsets. During each fold, four parts were trained, and the other part was validated. This was done five times so that each subset of the data was validated once, and this gives a stronger indication of the performance of the model.

In the first fold, the results indicate that the model attained an overall accuracy of 87%, which implies that it accurately classified most of the samples in each of the three classes. The Healthy class had the highest recall of up to, indicating that DenseNet121 was very sensitive to healthy cocoa leaves. Another class, Anthracnose, also performed well with a precision of 0.91, recall of 0.89 and an F1-score of 0.90, i.e. it was predicted reliably by the model. The CSSVD class however had the lowest scores, with a recall of 0.77 specifically, indicating that the model might have struggled to differentiate between CSSVD. Figures 4.3 and 4.4 show the confusion matrix and training and validation graphs respectively.

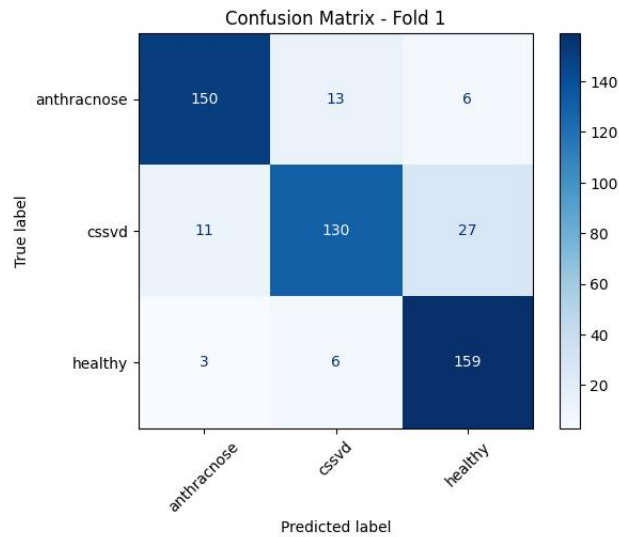


Figure 4.3: Confusion Matrix for the First Fold using DenseNet121

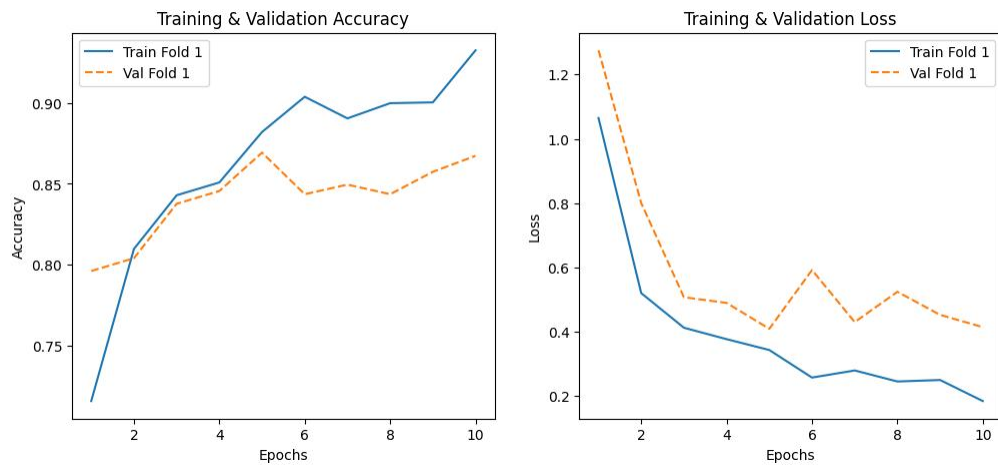


Figure 4.4: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 using DenseNet121

The overall accuracy achieved in the second fold was 83% meaning that the model performed well on all three classes. The highest recall of 0.89 belonged to the category of Anthracnose, which indicated that this model was especially capable of recognizing this disease. CSSVD and Healthy received a bit higher precision score, 0.84 and 0.85 respectively, but a bit lower recall score, which yielded F1-scores of 0.81 and 0.83. The overall performance was reasonably consistent with a macro and weighted average F1-score of 0.83. Figures 4.5 and 4.6 show the confusion matrix and training and validation graphs respectively.

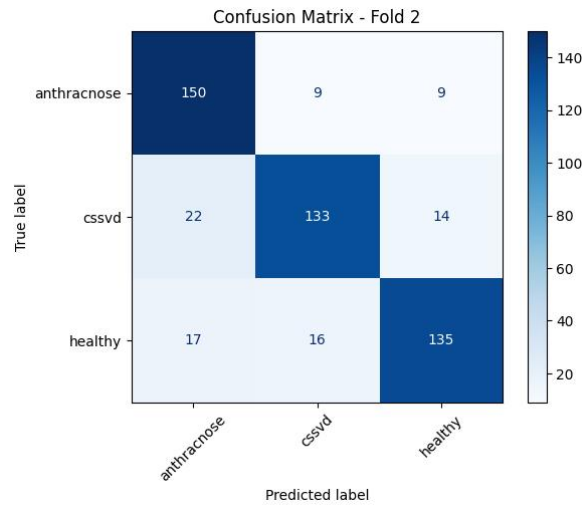


Figure 4.5: Confusion Matrix for the Second Fold using DenseNet121

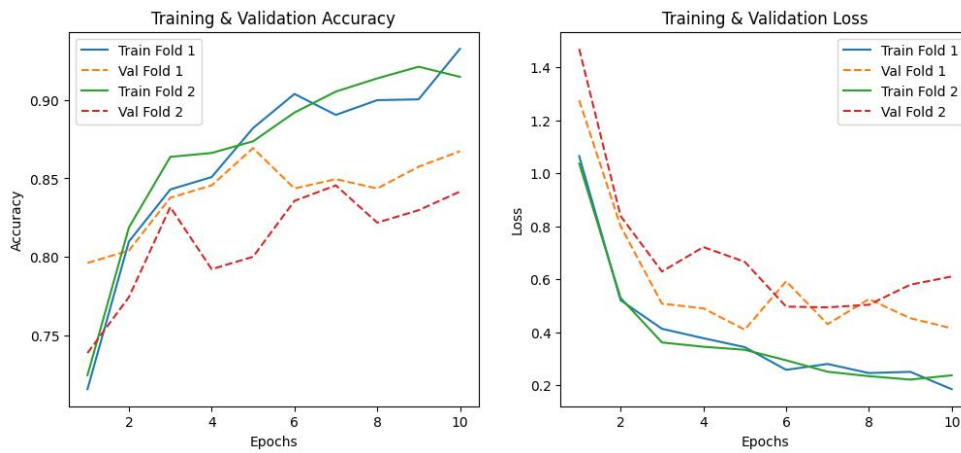


Figure 4.6: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 and Fold 2 using DenseNet121

As per the results on the third fold, the model attained an overall accuracy of 86% which shows that it also has performed well on all three classes. The Healthy class fared the best with a precision and recall of 0.88 achieving an F1-score of 0.88. Anthracnose also performed well with a recall of 0.88 and precision of 0.83, providing it with an F1-score of 0.86. The CSSVD class was close behind with an F1-score of 0.85, precision of 0.87 and recall of 0.83. The macro and weighted average F1-score was 0.86, which demonstrates that the model performed stably across the fold. Figures 4.7 and 4.8 show the confusion matrix and training and validation graphs respectively.

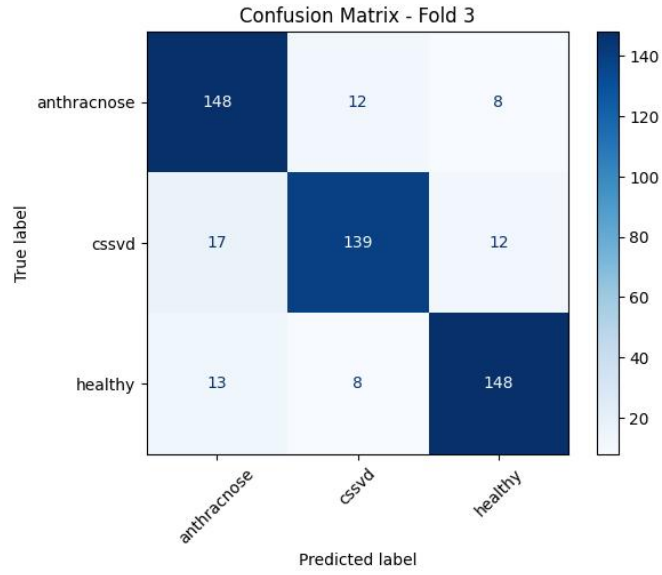


Figure 4.7: Confusion Matrix for the Third Fold using DenseNet121

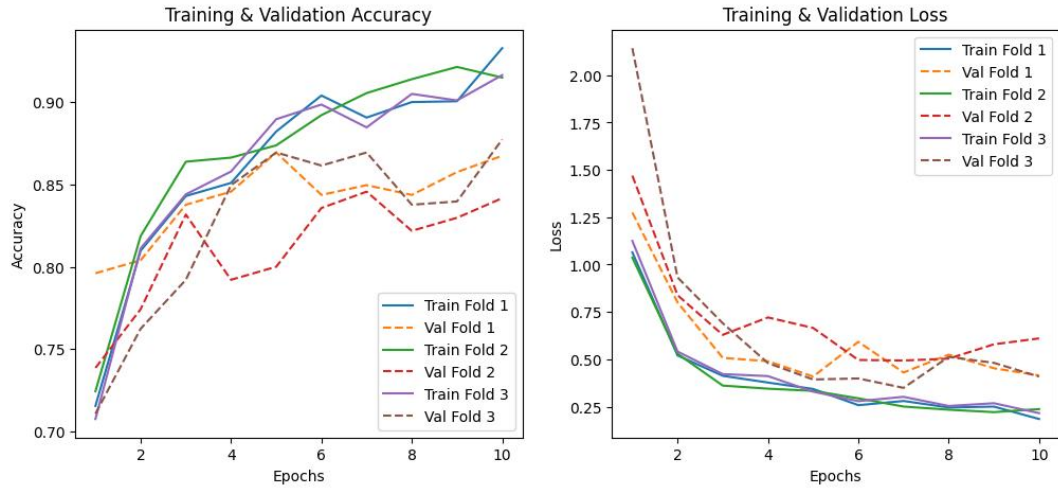


Figure 4.8: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 to Fold 3 using DenseNet121

The overall accuracy of the model in the fourth fold was 86% indicating a good performance on the three classes. The most successfully detected class was anthracnose with the highest recall of 0.93 and an F1-score of 0.87, although its precision was a little bit lower, 0.81. Although the CSSVD class achieved high precision of 0.90, its recall was lower at 0.77, indicating perhaps a few CSSVD cases were not detected easily, yielding an F1-score of 0.83. Both precision and recall were 0.86 and the F1-score was 0.86 in healthy samples, which were always well-classified. The macro and weighted average F1-scores were both 0.85, showing that

the model worked reasonably well on all classes. Figures 4.9 and 4.10 show the confusion matrix and training and validation graphs respectively.

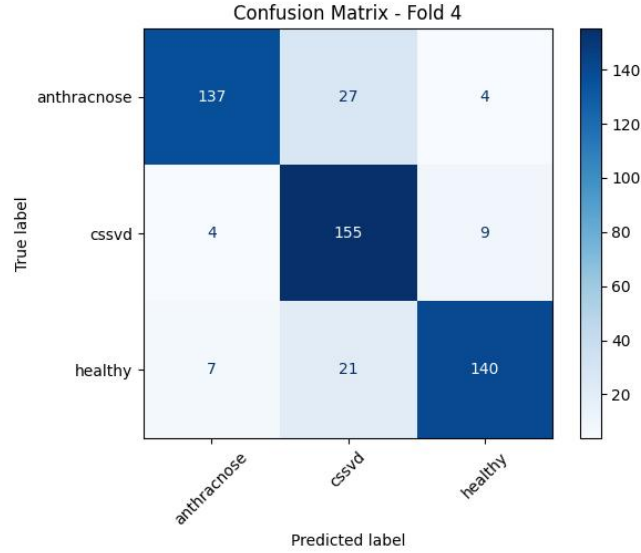


Figure 4.9: Confusion Matrix for the Fourth Fold using DenseNet121

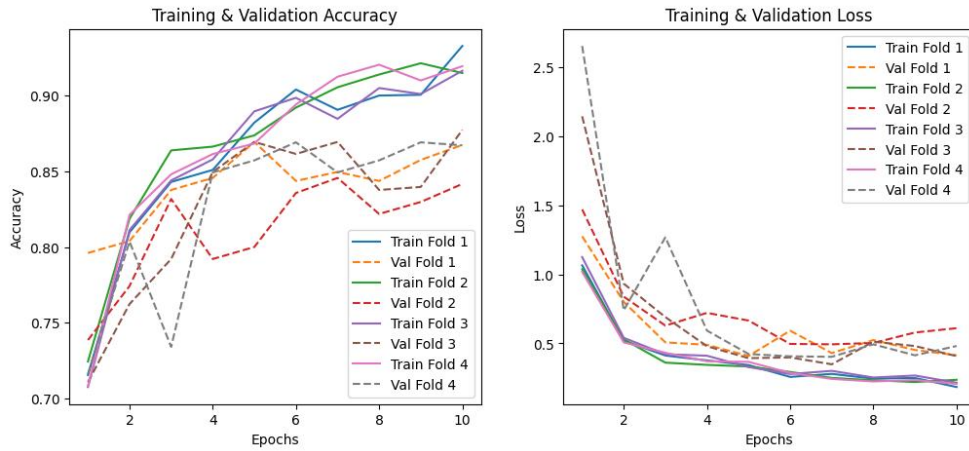


Figure 4.10: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 to Fold 4 using DenseNet121

The overall accuracy of the model, based on the evaluation of the fifth fold, was 86% with the confident classification in all three classes. The classes Healthy and Anthracnose obtained the same F1-score of 0.87, however, the class Healthy obtained high precision (0.92) and the class Anthracnose obtained better recall (0.82). CSSVD which had the least precision of 0.76 had the highest recall of 0.92 meaning that the model captured most cases of CSSVD but misclassified other classes as

CSSVD. Both the macro and weighted average F1-scores were 0.86 indicating a similar performance throughout the dataset. Figures 4.11 and 4.12 show the confusion matrix and training and validation graphs respectively.

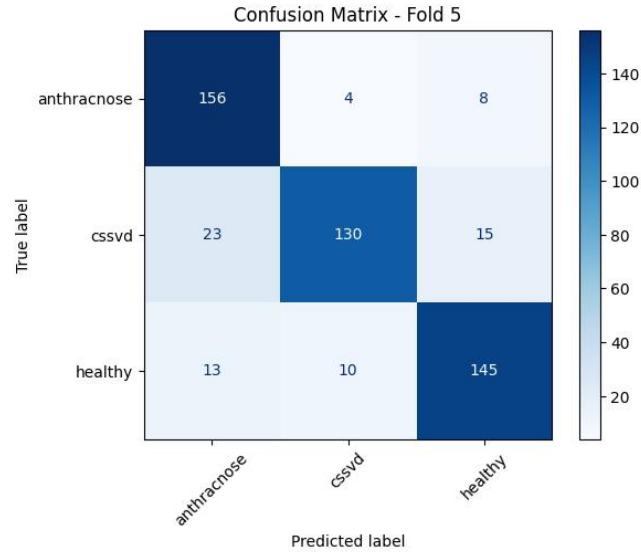


Figure 4.11: Confusion Matrix for the Fifth Fold using DenseNet121

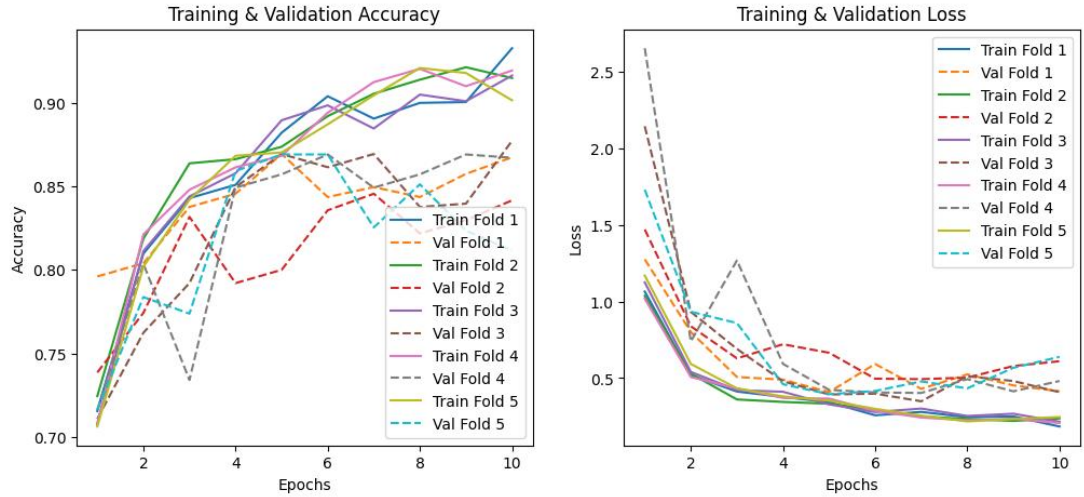


Figure 4.12: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 to Fold 5 using DenseNet121

Table 4.2 shows an overall summary of the evaluation results of all five folds using the 5-fold cross validation.

Table 4. 2: The Summary of the Results for Cross Validation on DenseNet-121

Class	Precision					Recall					F1-Score				
	1 st	2 nd	3 rd	4 th	5 th	1 st	2 nd	3 rd	4 th	5 th	1 st	2 nd	3 rd	4 th	5 th
Anthracnose	0.91	0.79	0.83	0.93	0.81	0.89	0.89	0.88	0.82	0.93	0.90	0.84	0.86	0.87	0.87
CSSVD	0.87	0.84	0.87	0.76	0.90	0.77	0.79	0.83	0.92	0.77	0.82	0.81	0.85	0.84	0.83
Healthy	0.83	0.85	0.88	0.92	0.86	0.95	0.80	0.88	0.83	0.86	0.88	0.83	0.88	0.87	0.86
Overall Metrics															
Accuracy											0.87	0.83	0.86	0.86	0.86
Macro avg	0.87	0.83	0.86	0.87	0.86	0.87	0.83	0.86	0.86	0.86	0.87	0.83	0.86	0.86	0.85
Weighted avg	0.87	0.83	0.86	0.87	0.86	0.87	0.83	0.86	0.86	0.86	0.87	0.83	0.86	0.86	0.85

Out of all the folds, the third fold performed in the most well-balanced and consistent manner and thus it is the best overall. It obtained the best validation accuracy of 87.7% and had low F1-scores across all the classes, 0.86 in Anthracnose, 0.85 in CSSVD, and 0.88 in Healthy. The validation loss was minimal at 0.3491 which shows that it learned well with a low tendency to overfit. Figure 4.13 shows the training and validation graphs.

4.3 EfficientNet-B0 Model Results

EfficientNetB0 is a convolutional neural network that uses compound scaling to scale all its parameters i.e. depth, width, and resolution of inputs equally depending on the predetermined coefficients. This technique enables the architecture to attain high precision on image classification without incurring a cost on computation and reduced parameters. EfficientNetB0 was used in this research to classify cocoa leaf diseases due to its light model nature and good generalization ability to learn and recognize signs of diseases in leaf images effectively. It is suitable in resource-limited environment makes it the most appropriate to integrate into a mobile-based plant disease detection system.

4.3.1 Training of EfficientNet-B0

The model had an accuracy of 88%. The best recall was observed in the Healthy class with a value of 0.93, which suggests consistent identification of healthy

samples. Anthracnose kept an even keel performance on precision and recall of 0.89. Nevertheless, the CSSVD class had a somewhat low recall of 0.83 indicating incorrect classification into the other classes, especially Anthracnose and Healthy. Table 4.3 shows the training results. Figures 4.14 and 4.15 show the confusion matrix and training and validation graphs respectively.

Table 4. 3: Summary of the Results for Transfer Learning on EfficientNetB0

Disease/Metric	Precision	Recall	F1-score
Anthracnose	0.89	0.88	0.89
CSSVD	0.89	0.83	0.86
Healthy	0.86	0.93	0.89
Accuracy			0.88
Macro Average	0.88	0.88	0.88
Weighted Average	0.88	0.88	0.88

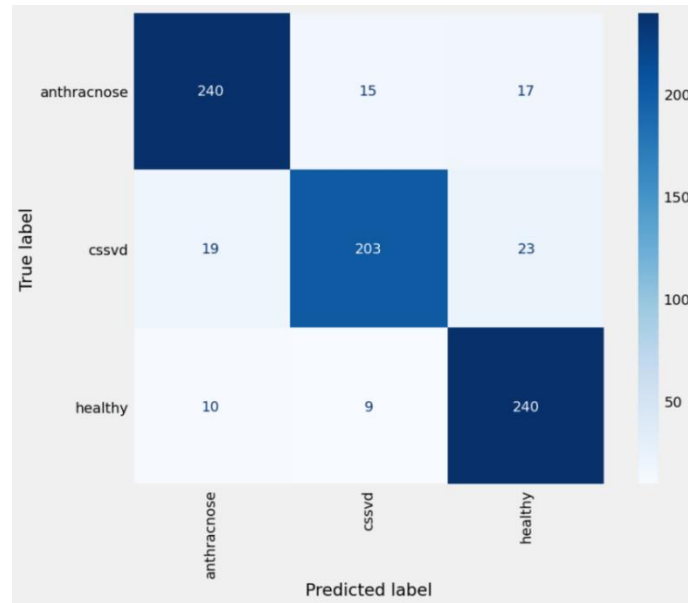


Figure 4.13: Confusion Matrix of Transfer Learning using EfficientB0

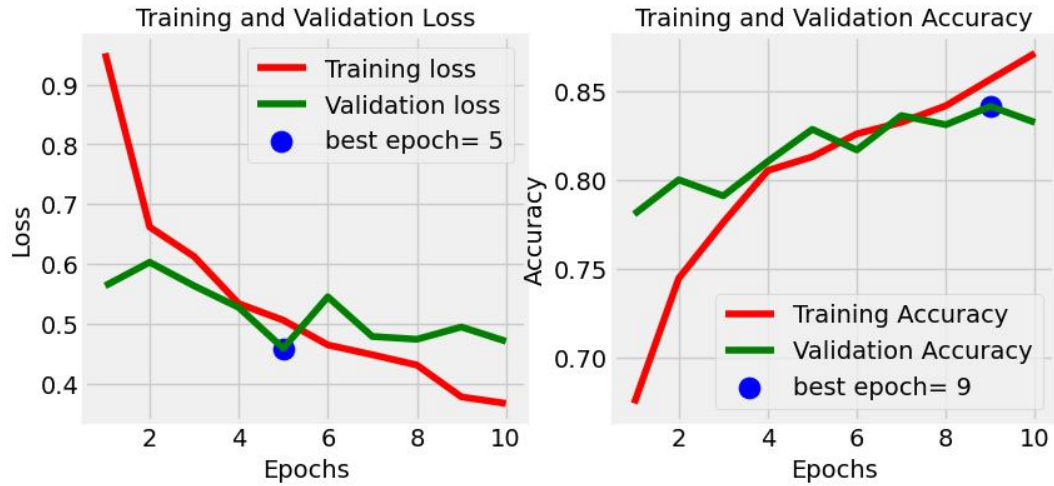


Figure 4.14: Graphs of Training Accuracy over Validation Loss and Training Loss over Validation Accuracy for Transfer Learning using EfficientB0

4.3.2 Evaluation of EfficientNetB0 Using 5-Fold Cross Validation

This section focuses on the performance of the Efficientnet-B0 architecture. The model was also trained on the five-fold cross-validation method.

On the first fold, the model had a validation accuracy of 86%. Healthy was reliably identified with the highest classification performance recall of 0.92 and F1-score of 0.89 which is a strong measure of sensitivity, among the classes. This means that anthracnose had high recall (0.89) and a good F1-score 0.86 representing capability to detect it well. Nevertheless, CSSVD was comparatively less sensitive i.e., recall of 0.76, indicating that there was a misclassification of infected samples. However, the model performed well on a uniform basis in all classes with a regular macro-averaged F1-score as 0.86. Figures 4.16 and 4.17 show the confusion matrix and training and validation graphs respectively.

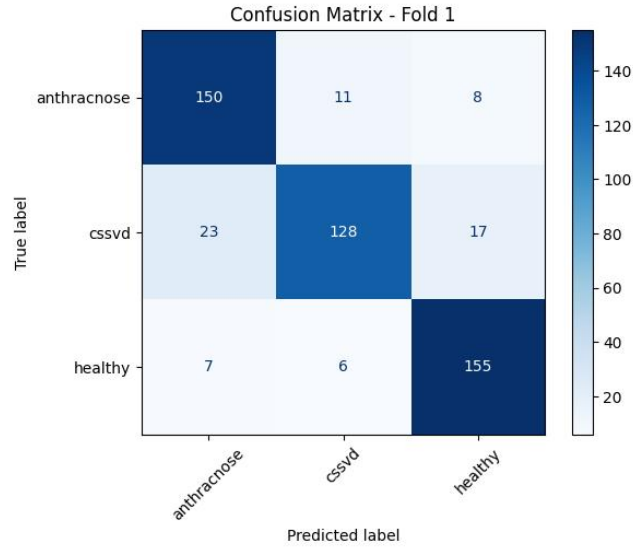


Figure 4.15: Confusion Matrix of Transfer Learning using EfficientNetB0

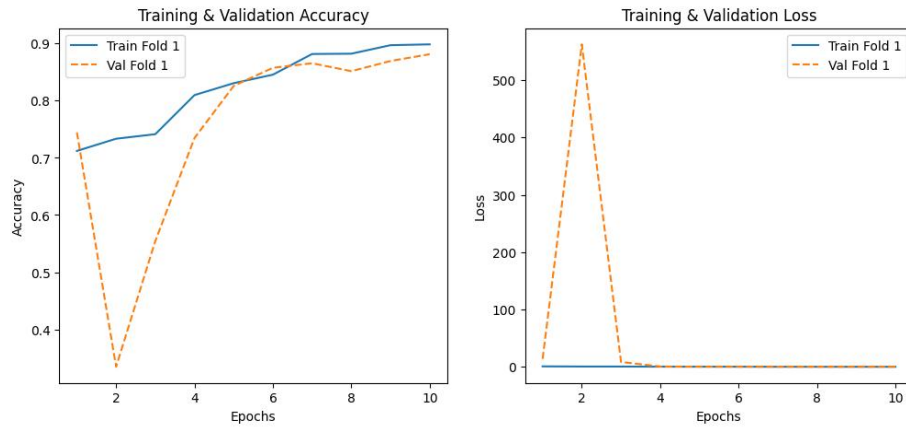


Figure 4.16: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 using EfficientNetB0

With the second fold, the overall accuracy of the model on the validation set was 86%, which is exhibited by proper classification in the three classes. Anthracnose was the most precise with 0.92 showing high level of confidence when predicting this class, whereas its recall is a bit low at 0.83. CSSVD performed the most equal and presented the results precision 0.85 and recall 0.87. The healthy samples were also appropriately classified recording a recall of 0.89. The F1-scores indicate that there is a consistent performance of the model in its capacity to generalize across the categories, which is that the macro-averaged value of 0.86. Figures 4.18 and 4.19 show the confusion matrix and training and validation graphs respectively.

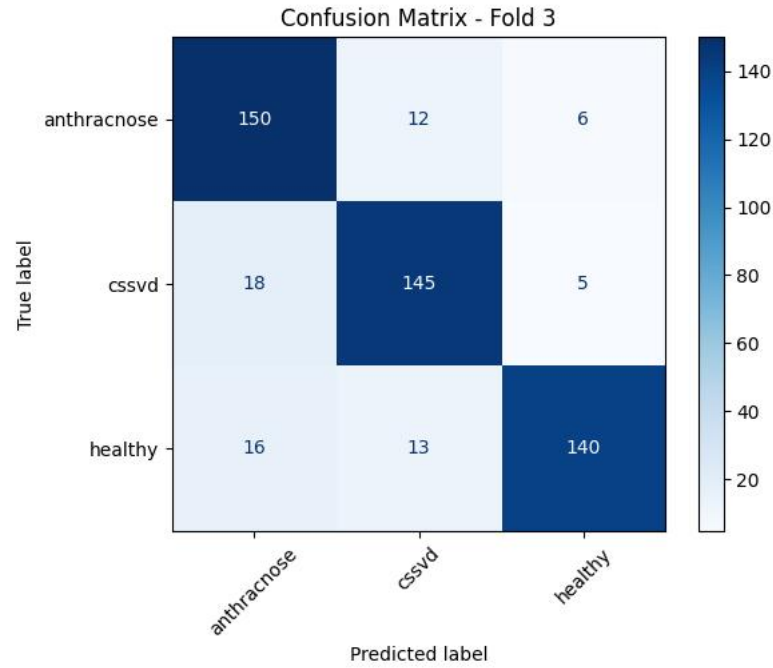


Figure 4.17: Confusion Matrix of Transfer Learning using EfficientNetB0

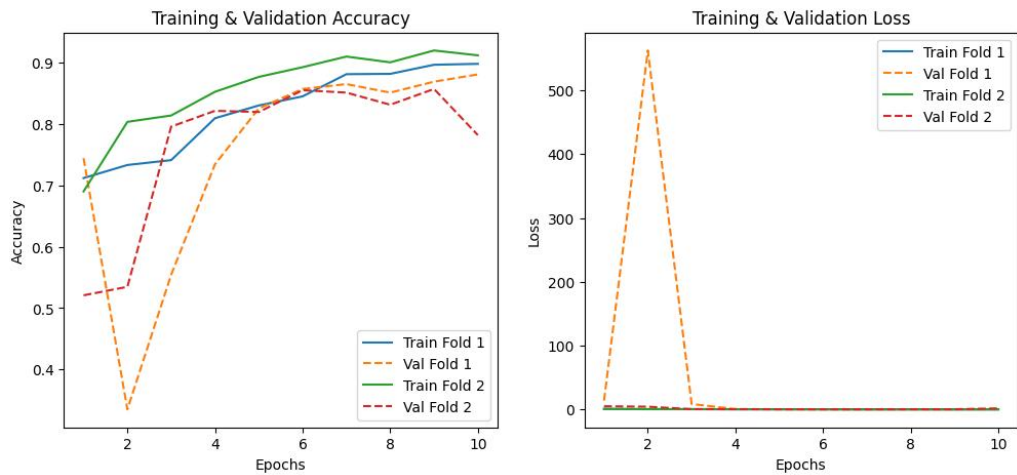


Figure 4.18: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 to Fold 2 using EfficientNetB0

The third fold resulted in the validation accuracy of 86% and the balanced output in all classes of the classification. Precision was highest in healthy samples, that is, 0.93 and suggests high confidence in the predictions of this class, but there was a slightly lower recall of 0.83. Anthracnose has the highest recall of 0.89 indicating that the

combination could detect most of the infected samples, whereas CSSVD had a stable balance with a recall of around 0.85 and 0.86 and precisions. Stable and generalizable performance of the model is proved along with the macro and weighted F1-scores of 0.86. Figures 4.20 and 4.21 show the confusion matrix and training and validation graphs respectively.

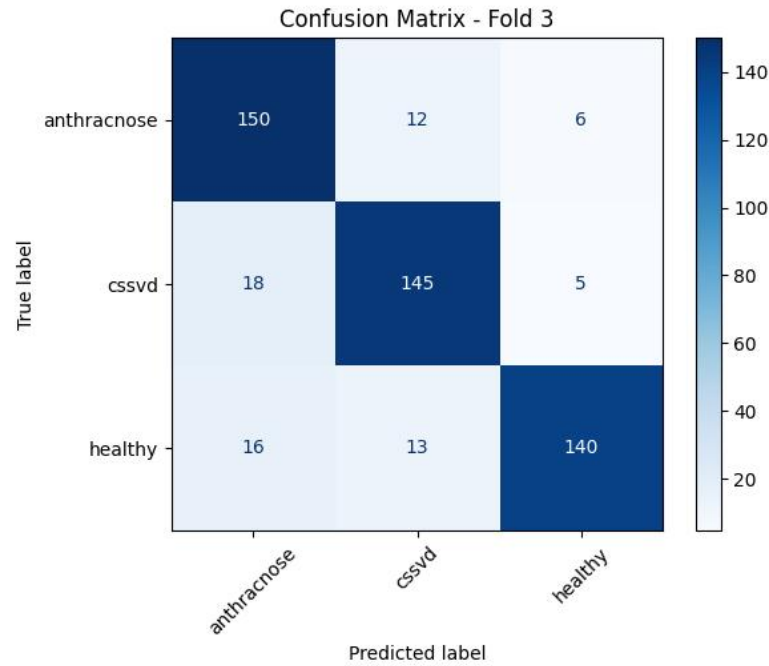


Figure 4.19: Confusion Matrix for the Third Fold using EfficientNetB0

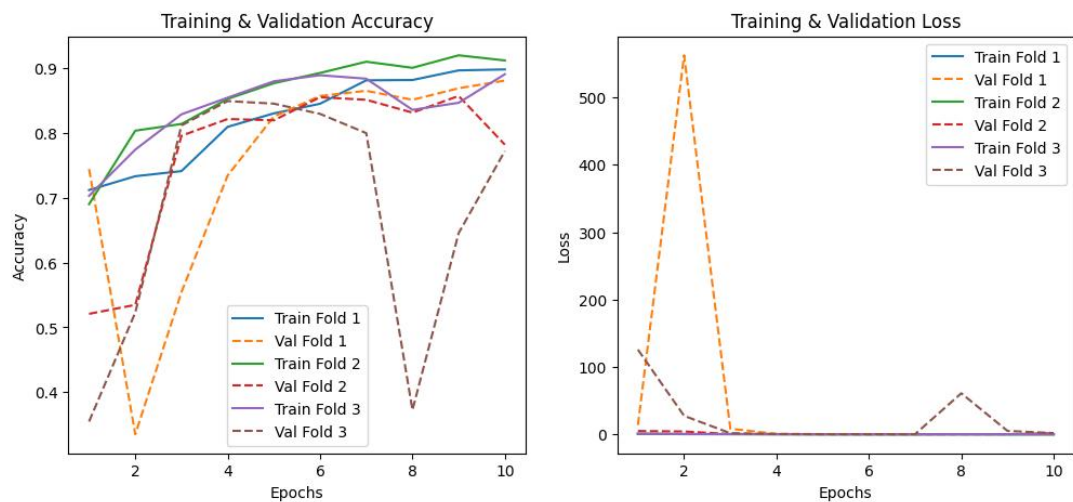


Figure 4.20: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 to Fold 3 using EfficientNetB0

The overall validation accuracy of the model in the fourth fold was 85%, and it was comparatively balanced when it came to classes. Anthracnose had the highest precision rate of 0.94 which means that it strongly believes in its predictions of that class, but because of the low value of the recall of 0.81, some of the true cases were overlooked. The recall of healthy samples was found to be 0.92 indicating that the model was quite good with classifying healthy leaves, but its precision is lower 0.81, which indicates that there may be false positives. CSVVD demonstrated moderate results with 0.83 precisions and 0.83 recalls. These are proven by the macro and weighted F1-scores of 0.85 which verify stability in all classes. Figures 4.22 and 4.23 show the confusion matrix and training and validation graphs respectively.

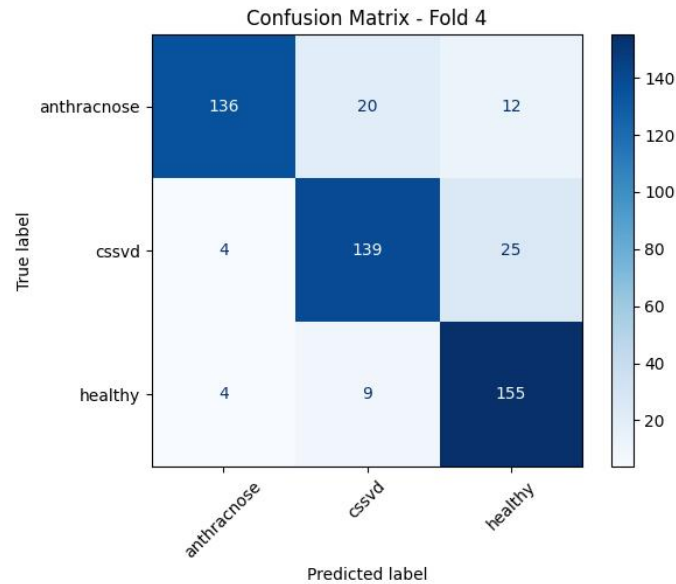


Figure 4.21: Confusion Matrix for the Fourth Fold using EfficientNetB0

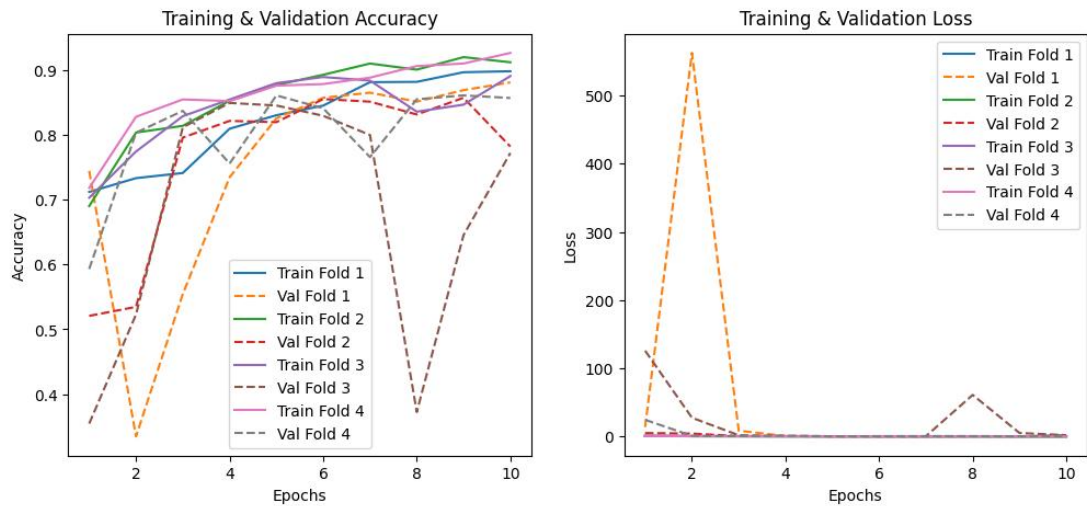


Figure 4.22: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 to Fold 4 using EfficientNetB0

In the final fold, 87% accuracy was achieved indicating a balanced and good performance. Anthracnose reached the farthest F-1 score of 0.90, which means that both precision (0.92) and recall (0.88) were high, meaning consistent and reliable predictions. Healthy samples appeared at the most apt with the pinnacle recall of 0.96, which infers that most of the healthy leaves were precisely categorized, yet with its lower exactness of 0.80, implying that there were some false positives. CSSVD was very specific with 0.92 but not as good as recall with 0.79 which reflects the tendency to overlook several real CSSVD cases. The macro-mean of the F1-score is 0.87, which ensures the high robustness of the model generalization to all the classes. Figures 4.24 and 4.25 show the confusion matrix and training and validation graphs respectively.

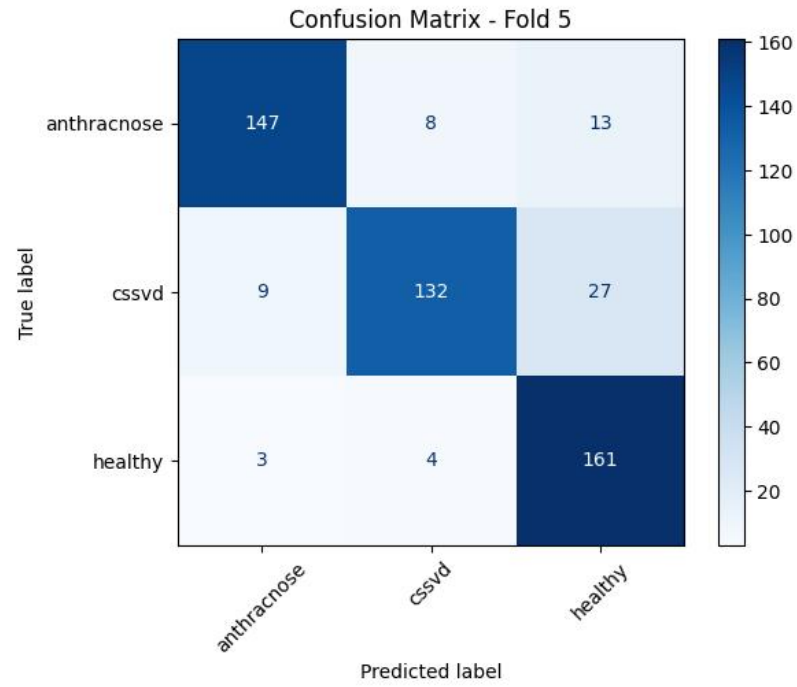


Figure 4.23: Confusion Matrix for the Fourth Fold using ResNet50

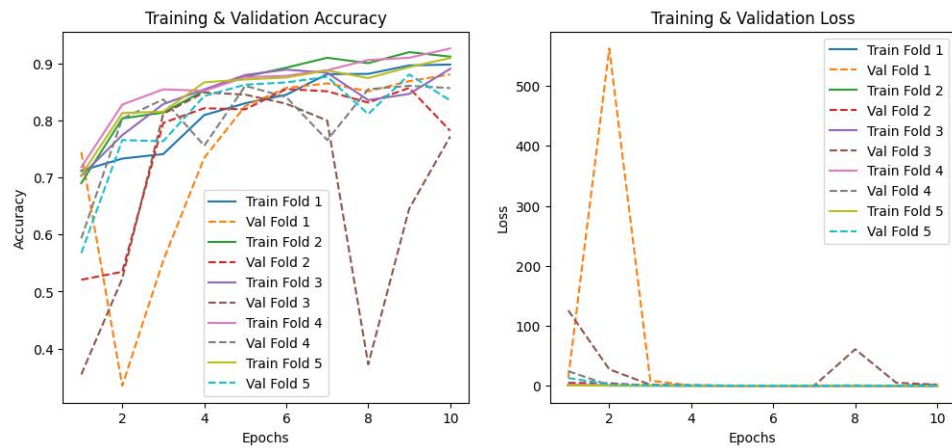


Figure 4.24: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 to Fold 5 using EfficientNetB0

Table 4.4 shows an overall summary of the evaluation results of all five folds using the 5-fold cross validation.

Table 4. 4: Summary of the Results for Cross Validation on EfficientNetB0

Class	Precision					Recall					F1-Score				
	1 st	2 nd	3 rd	4 th	5 th	1 st	2 nd	3 rd	4 th	5 th	1 st	2 nd	3 rd	4 th	5 th
Anthracnose	0.88	0.86	0.87	0.90	0.89	0.87	0.88	0.86	0.88	0.87	0.87	0.87	0.86	0.89	0.88
CSSVD	0.84	0.85	0.84	0.82	0.86	0.80	0.83	0.81	0.83	0.80	0.82	0.84	0.82	0.82	0.83
Healthy	0.85	0.86	0.86	0.88	0.87	0.91	0.88	0.89	0.90	0.90	0.88	0.87	0.87	0.89	0.88
Overall Metrics															
Accuracy											0.86	0.86	0.86	0.86	0.86
Macro avg	0.86	0.87	0.87	0.86	0.88	0.86	0.86	0.86	0.85	0.87	0.86	0.86	0.86	0.85	0.86
Weighted avg	0.86	0.87	0.87	0.86	0.88	0.86	0.86	0.86	0.85	0.87	0.86	0.86	0.86	0.85	0.86

4.4 ResNet-50 Model

ResNet incorporates shortcut (or residual) connections in which the input of one layer can skip over some layers and be added to the output. This has been shown to help retain important features when training and helps solve the issue of the vanishing gradient allowing deeper networks to be trained.

4.4.1 Training of ResNet-50

The model achieved an overall accuracy of 88%, with the Healthy class having the highest recall (0.94), meaning the high capability of identifying healthy samples. Nevertheless, it is less precise (0.83) indicating that a certain number of diseased leaves were classified as Healthy. Anthracnose revealed the best precision (0.94) and high recall (0.90) whereas CSSVD exhibited a balanced performance with precision of 0.89 and recall of 0.83. Table 4.3 shows the training results. Figures 4.26 and 4.27 show the confusion matrix and training and validation graphs respectively.

Table 4. 5 : The Summary of the Results for Transfer Learning using Resnet-50

Disease/Metric	Precision	Recall	F1-score
Anthracnose	0.94	0.90	0.92
CSSVD	0.89	0.83	0.86

Disease/Metric	Precision	Recall	F1-score
Healthy	0.83	0.94	0.88
Accuracy			0.88
Macro Average	0.89	0.89	0.89
Weighted Average	0.89	0.88	0.88

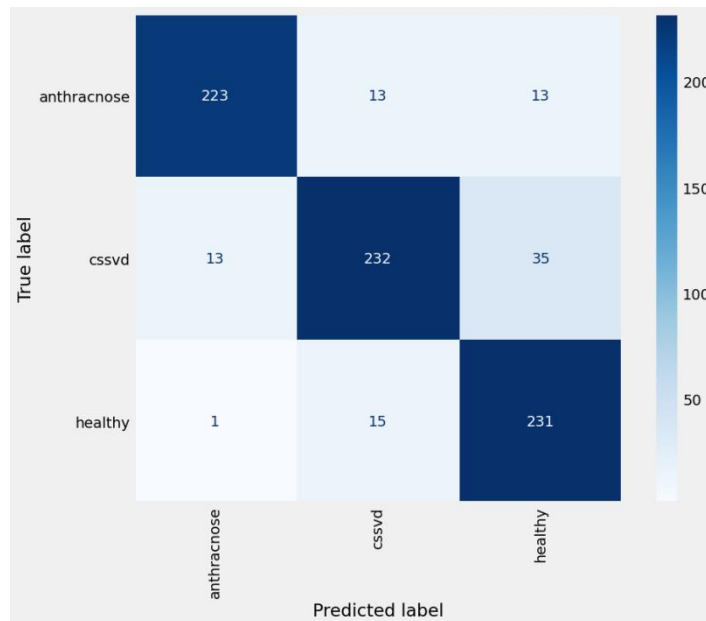


Figure 4.25: Confusion Matrix of Transfer Learning on ResNet50



Figure 4.26: Graphs of Training over Validation Loss and Accuracy of ResNet50

4.4.2 Evaluation of ResNet-50 Using 5-Fold Cross Validation

For the first fold, the model showed reasonable classification results on the three classes with an overall accuracy of 86%. The class with the lowest recall was CSSVD with a value of 0.76, which means that this model struggled to recognize this disease sometimes. Conversely, Healthy samples had the best recall of 0.92 with an F1-score of 0.89, indicating high sensitivity and characterized stable classification. Anthracnose was also identified with reasonable accuracy achieving a precision of 0.83 and a recall of 0.89, combining to give an F1-score of 0.86. The macro and weighted average of precision, recall and F1-score were all 0.86-0.87, which showed that the model has good generalization ability across all the classes with minimal bias. Figures 4.28 and 4.29 show the confusion matrix and training and validation graphs respectively.

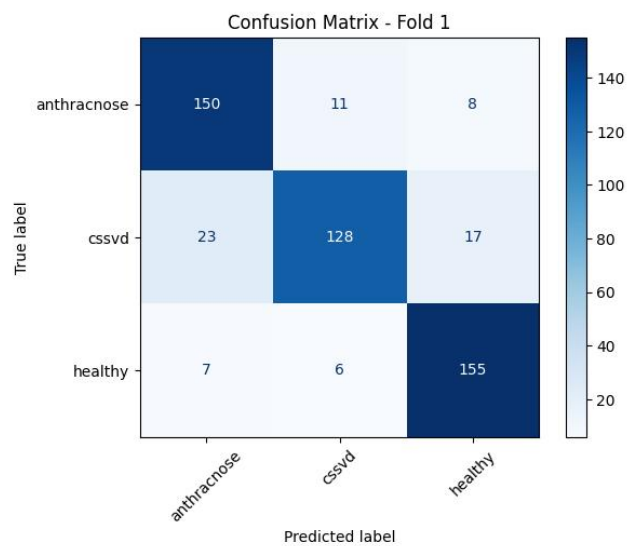


Figure 4.27: Confusion Matrix for the First Fold using ResNet50

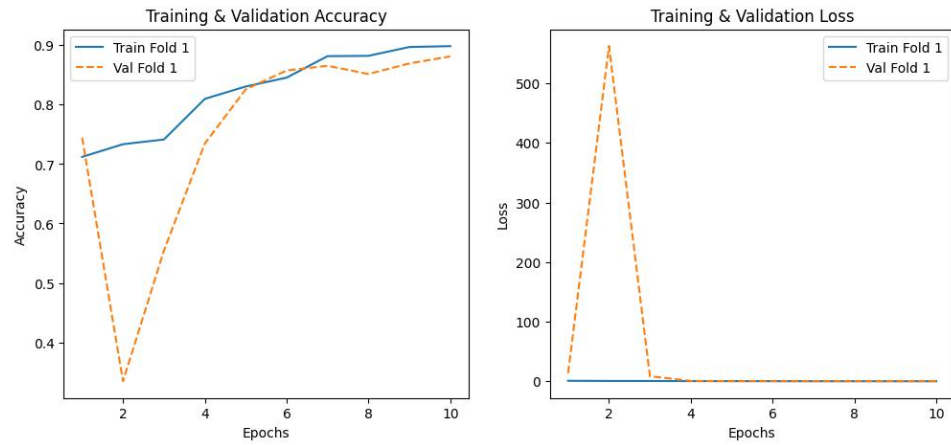


Figure 4.28: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 using ResNet50

On the second fold, the model got an overall accuracy of 86% and relatively balanced accuracy on all classes. Anthracnose had the best precision of 0.92 meaning that the model is very confident with its prediction of this class, but the recall was a little bit lower, 0.83. The CSSVD class had the best recall of 0.87 and a good F1-score of 0.86 that indicated a reasonable balance between sensitivity and precision. Likewise, the Healthy class had a recall of 0.89 and precision of 0.83, which gives an F1-score of 0.86. Both macro and weighted average of precision, recall and F1-score were similar (0.86-0.87), which indicated the overall strength of the model and its generalization capacity across all three classes with a low bias. Figures 4.30 and 4.31 show the confusion matrix and training and validation graphs respectively.

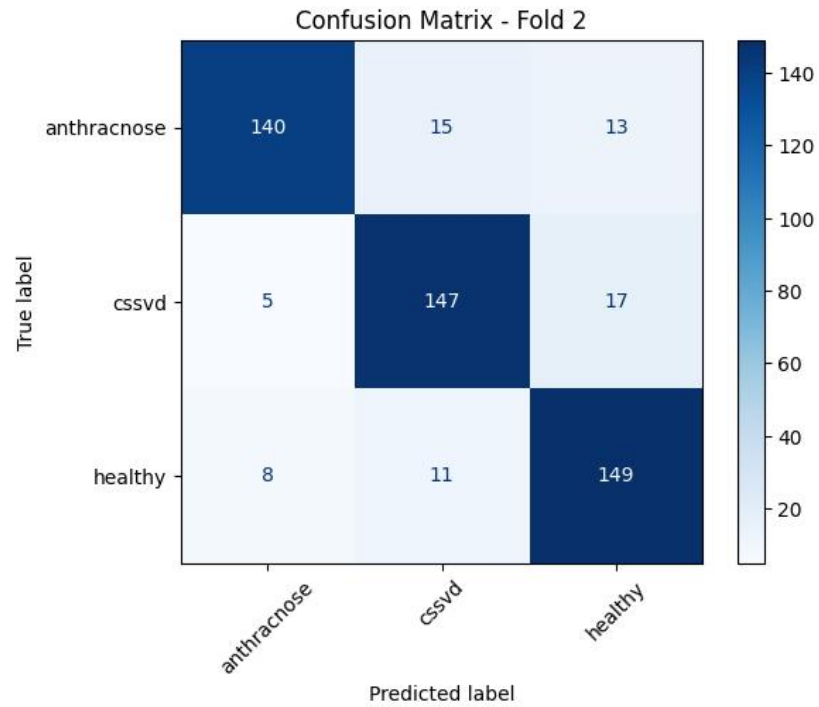


Figure 4.29: Confusion Matrix for the Second Fold using ResNet50

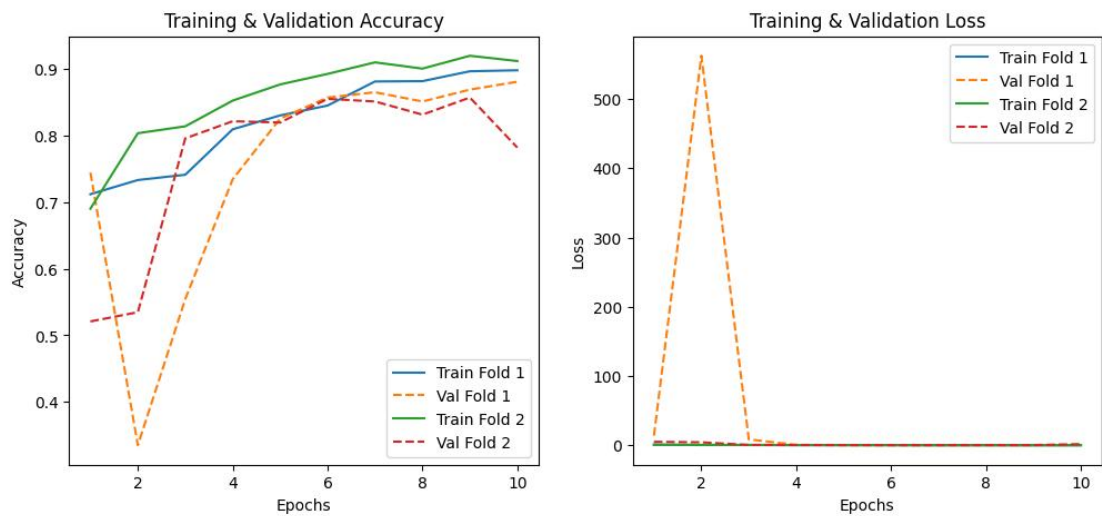


Figure 4.30: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 to Fold 2 using ResNet50

The model showed consistency in the classification accuracy on all the three classes on the third fold with an overall accuracy of 86%. The Healthy class had the best precision of 0.93 which means that the model is very confident in predicting healthy sample, though the recall was a bit lower, 0.83. Conversely, the Anthracnose class

got a recall of 0.89, indicating that the model was very sensitive to identify cases of anthracnose, however, its precision was relatively lower at 0.82. The CSSVD class was more evenly balanced in its performance with both precision and recall very close at 0.85 and 0.86 respectively giving a respectable F1-score of 0.86. The macro and weighted averages of all the metrics were similar 0.86-0.87 which indicates that the model was consistent throughout the data without being biased on any class. Figures 4.32 and 4.33 show the confusion matrix and training and validation graphs respectively.

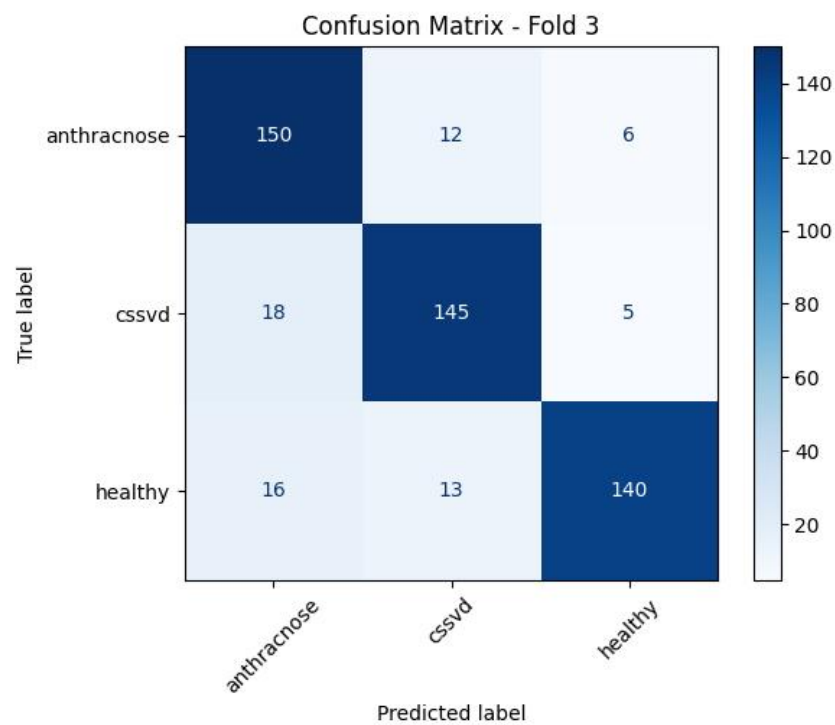


Figure 4.31: Confusion Matrix for the Third Fold using ResNet50

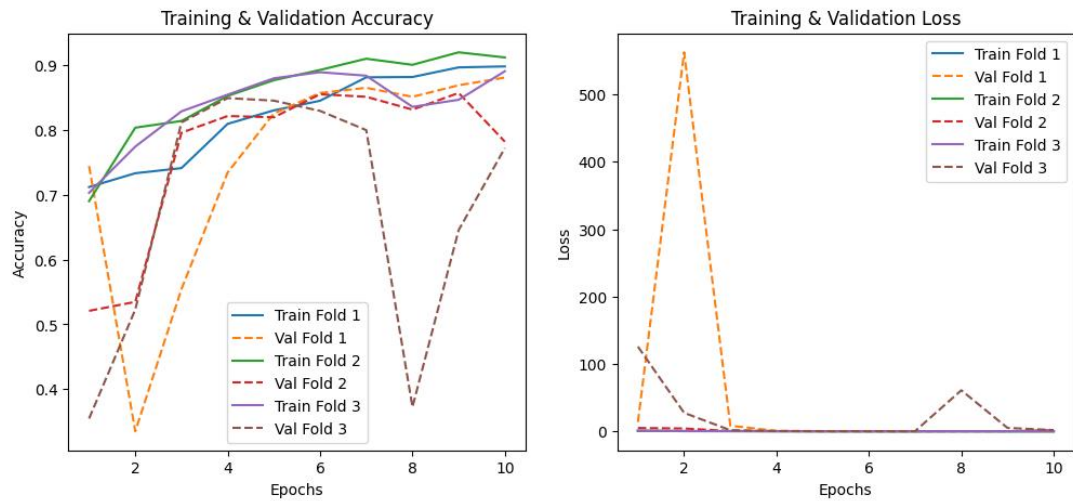


Figure 4.32: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 to Fold 3 using ResNet50

The model has achieved a general accuracy of 85% and shows the consistency in the prediction across all three classes. The Anthracnose class had the best precision of 0.94 which means that the model was highly accurate in predicting anthracnose samples, but its recall was comparatively low at 0.81. On the contrary, the Healthy class had the highest recall of 0.92, indicating that the model was sensitive to identify healthy leaves, however, the precision was a bit lower, 0.81. The class CSSVD presented a well-balanced profile, with precision and recall equal to 0.83, which resulted in the balanced F1-score equal to 0.83. The macro and weighted average of all measures was 0.85 0.86, which indicated the overall balanced and trustworthy classification ability of the model. Figures 4.34 and 4.35 show the confusion matrix and training and validation graphs respectively.

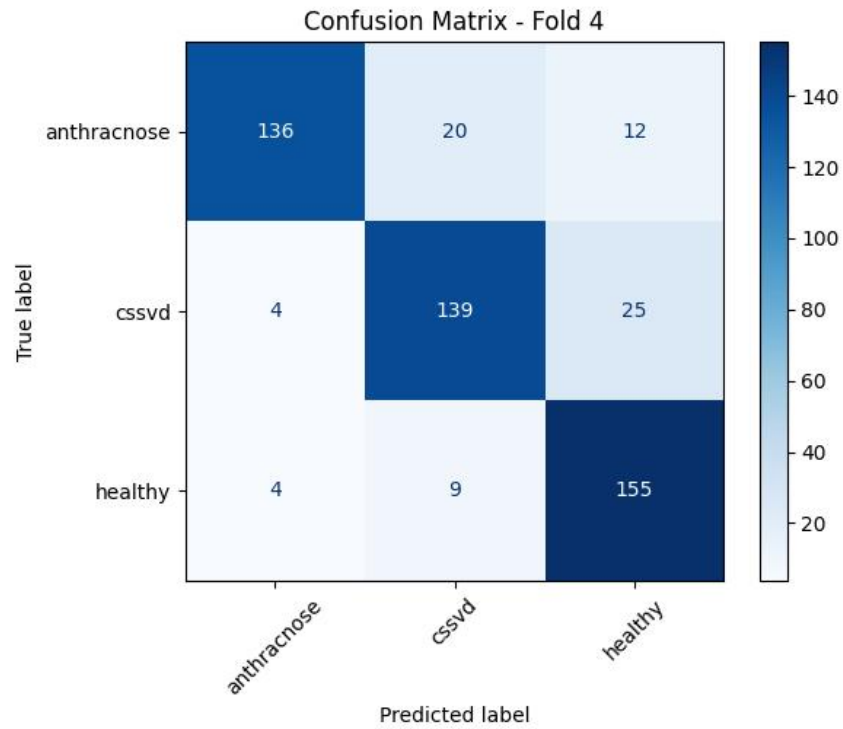


Figure 4.33: Confusion Matrix for the Fourth Fold using ResNet50

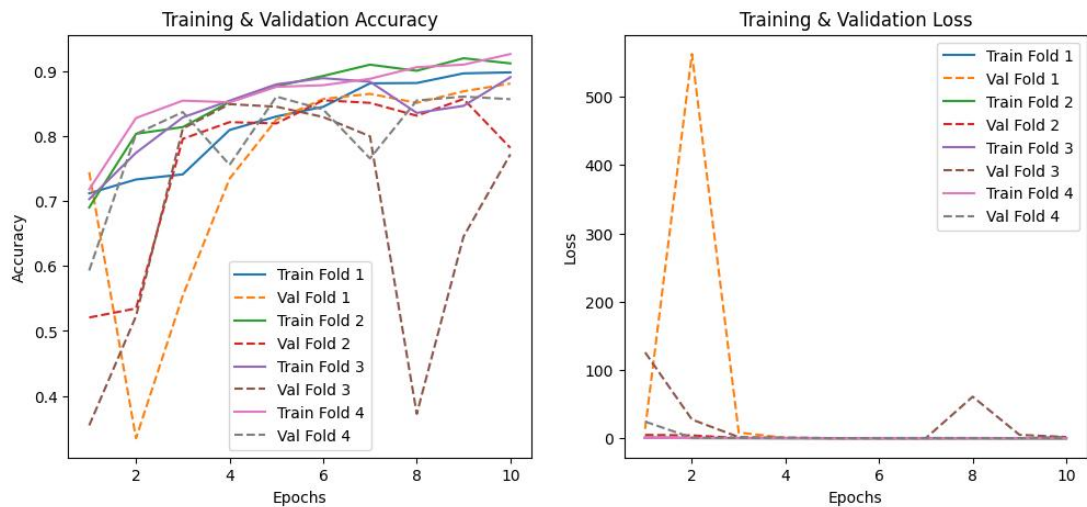


Figure 4.34: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 to Fold 4 using ResNet50

The model attained an overall accuracy of 87.02%, and all the three classes were well represented. Anthracnose had the best F1-score of 0.90 which was backed by a precision of 0.92 and a recall of 0.88, showing that the model was effective in predicting cases of anthracnose. The precision was equally high at 0.92 in classifying CSSVD samples, but the recall was slightly lower at 0.79, indicating that a few

samples might have been misclassified. The Healthy class recorded the best recall value of 0.96 demonstrating the ability of the model to identify healthy leaves although the precision value was 0.80. Both the macro and weighted average of precision, recall, and F1-score were equally consistent at around 0.87 0.88, indicative of a balanced performance and model generalizability on the dataset. Figures 4.36 and 4.37 show the confusion matrix and training and validation graphs respectively. Table 4.5 shows an overall summary of the evaluation results of all five folds using the 5-fold cross validation.

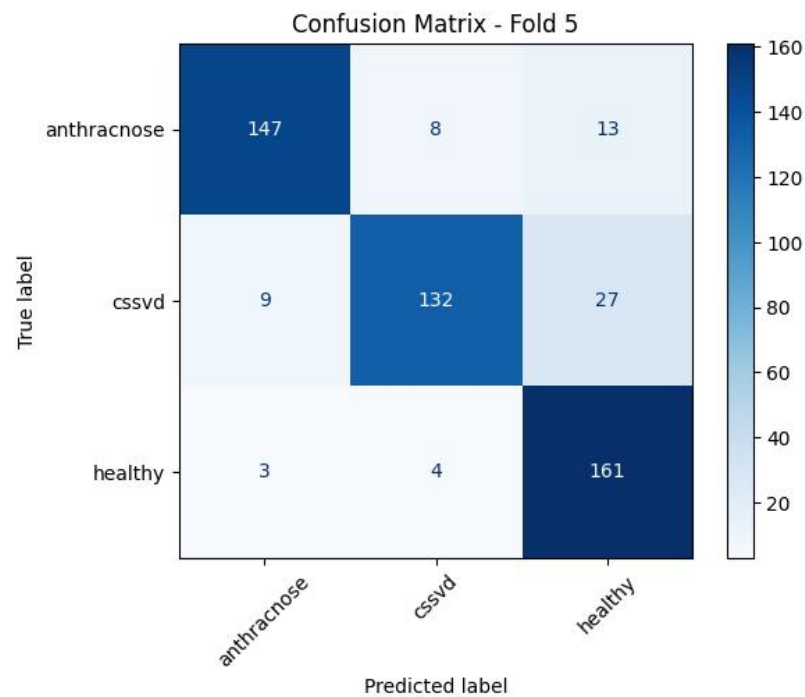


Figure 4.35: Confusion Matrix for the Fourth Fold using ResNet50

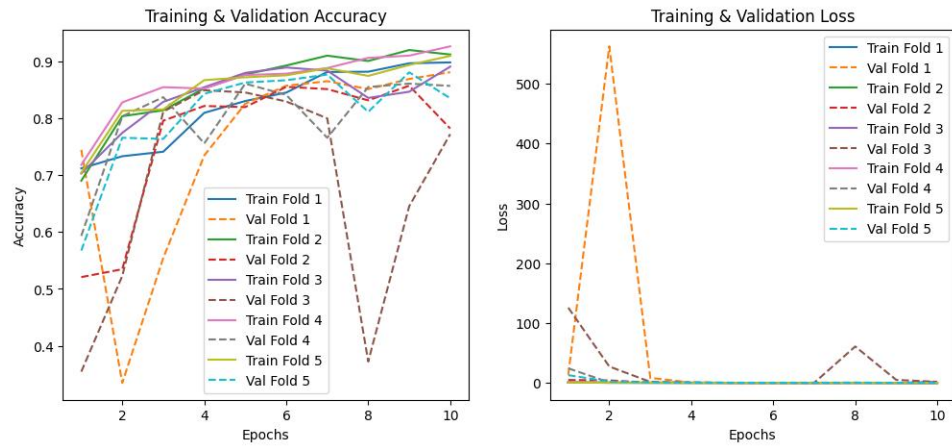


Figure 4.36: Graphs of Training Accuracy over Validation Accuracy and Training Loss over Validation Loss for Fold 1 to Fold 5 using ResNet50

Table 4. 6: The Summary of the Results for Cross Validation using Resnet-50

Class	Precision					Recall					F1-Score				
	1 st	2 nd	3 rd	4 th	5 th	1 st	2 nd	3 rd	4 th	5 th	1 st	2 nd	3 rd	4 th	5 th
Anthracnose	0.83	0.92	0.82	0.94	0.92	0.89	0.83	0.89	0.81	0.88	0.86	0.87	0.85	0.87	0.90
CSSVD	0.88	0.85	0.85	0.83	0.92	0.76	0.87	0.86	0.83	0.79	0.82	0.86	0.86	0.83	0.85
Healthy	0.86	0.83	0.93	0.81	0.80	0.92	0.89	0.83	0.92	0.96	0.89	0.86	0.87	0.86	0.87
Overall Metrics															
Accuracy											0.86	0.86	0.86	0.85	0.87
Macro avg	0.86	0.87	0.87	0.86	0.88	0.86	0.86	0.86	0.85	0.87	0.86	0.86	0.86	0.85	0.87
Weighted avg	0.86	0.87	0.87	0.86	0.88	0.86	0.86	0.86	0.85	0.87	0.86	0.86	0.86	0.85	0.87

Among all the folds, the fifth fold worked the best-balanced and consistent way and hence it is the best overall. It attained the best validation accuracy of 88.1% and its F1-scores were high on all the classes- 0.90 in Anthracnose, 0.85 in CSSVD, and 0.87 in Healthy. The model training loss was 0.3561, which is low thereby demonstrating that the model learns well with minimal overfitting.

The fold stability improved gradually during training, with an initial low accuracy of 63.1% and high loss of 1.4231 in the initial epoch but improved tremendously after epoch 9. It is worth noting that the model achieved the best performance in epoch 9 and then slightly dropped in the last epoch. Such consistency and high generalization on all classes makes Fold 5 the best and most strong and reliable fold overall.

4.5 Overall Comparative Analysis of Trained Models

Table 4.7 gives an overall analysis of the results on all three models and the evaluation with cross validation

Table 4. 7: Table showing an overall comparison of all trained models and evaluation

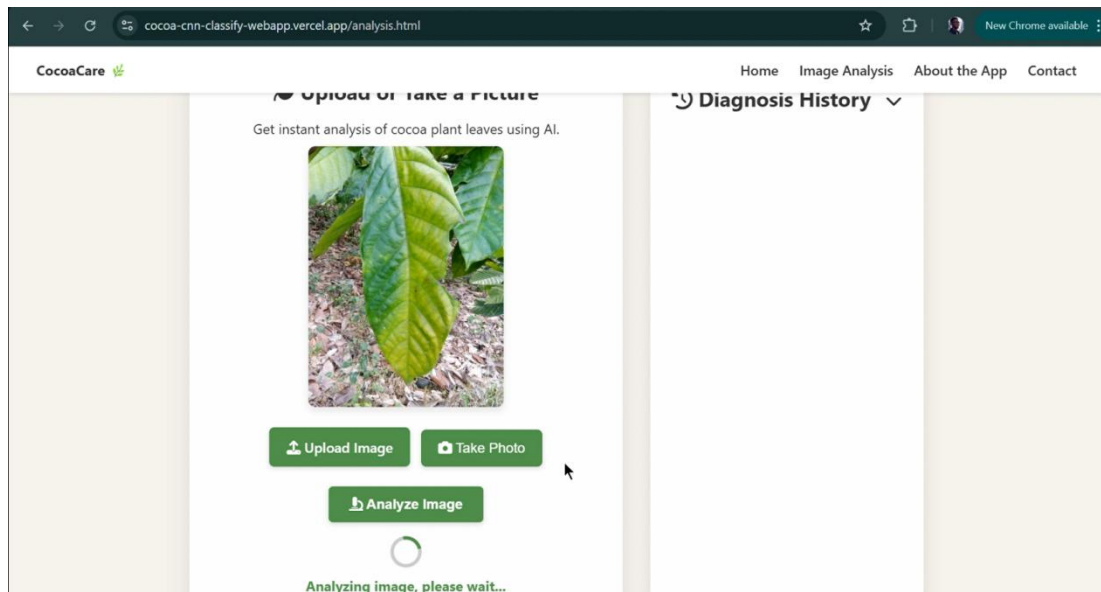
	Transfer Learning	Cross Validation
DenseNet121	90%	87%
EfficientNetB0	88%	86%
ResNet50	88%	87%

4.6 Web Application Deployment

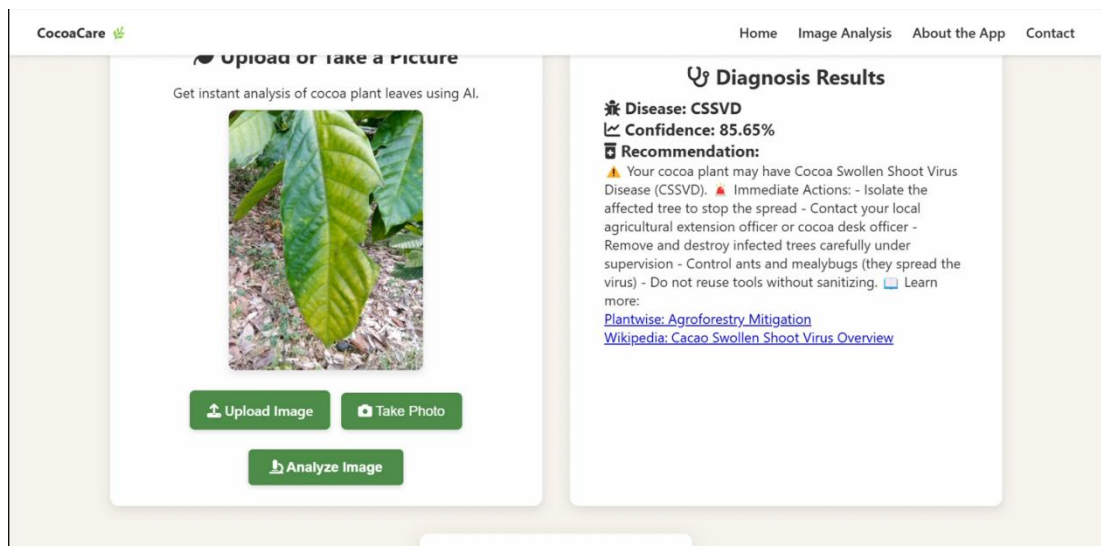
The web application has been tested on both mobile and desktop browser versions of Chrome. Essentially, the hosted frontend on Vercel made successful connections with the backend Flask API that rested on ngrok on Kaggle. Users could send pictures, view them as well as predict in an average of 4 to 6 seconds on fast internet, and up to a minute on slow internet conditions. There were predictions of the disease class, a confidence score, and context-specific recommendations.

Other functions including switching between dark/light theme, diagnosis history as well as mobile responsiveness were working sufficiently as intended. The feature of diagnosis history made it possible to see the previous 10 predictions including pictures, time stamps, and recommendations as it was more convenient to use the app.

Although the application showed steady performance in controlled settings with access to the internet, it is agreed that ngrok tunnels can last only a small amount of uptime and must be reinitialized, and thus it is not recommended to use it in the long run when the app is meant to be publicly hosted without premium or other alternatives.



(a)



(b)

Figure 4.37: The deployed model (a) analysing an image of an infected cocoa plant and (b) showing results.

4.7 Evaluation of Model Performance on Web Application

After deployment, A sample of 30 leaf images with known labels were uploaded through the web application. The deployed model accurately classified 27 out of 30 samples, yielding an accuracy of 90% aligning closely with accuracy shown during

offline testing. Overall, the deployment proved effective for real-world, user-interactive scenarios.

Table 4. 8: Results of testing the deployed model.

Class	No. Of Correct Predictions	Accuracy (%)
Anthracnose	9	90
CSSVD	8	80
Healthy	10	100

4.8 Chapter Summary

This chapter showed the performance and results of the three trained Convolutional Neural Network (CNN) models, which are DenseNet121, ResNet50 and EfficientNetB0 in the classification of cocoa leaf diseases. The fine-tuning of the models was done on a balanced dataset via transfer learning and the main metrics of evaluation were an accuracy, precision, recall, and F1-score.

DenseNet121 had the best performance (according to test accuracy of 90%) and balanced performance amongst the three disease classes of (Healthy, CSSVD, Anthracnose) out of the three models. Class-wise assessment showed that DenseNet121 performed very well specially in terms of recall of CSSVD as it is essential in early and accurate detection.

The most effective model was implemented in a web application that runs in real time, and the functionality aspect was measured on the basis of responsiveness, excellent user experience, and user-friendly applicability. The application was able to enable users to post or upload leaf images, and the user could get AI-based prediction with informative recommendation.

CHAPTER FIVE

SUMMARY, RECOMMENDATIONS AND CONCLUSIONS

5.1 Summary of Findings

The research comprised the design and implementation of a classification model that used convolutional neural networks to help in detecting cocoa leaf diseases with regard to Cocoa Swollen Shoot Virus Disease (CSSVD) and Anthracnose. The study dealt with these issues arisen by traditional methods of detecting the diseases in cocoa farming, which include inaccurate and sluggish nature besides requiring the expertise of people.

More than 9,000 images in KaraAgro AI Cocoa Dataset were available to train the model. In order to balance it, 2,584 images in each of the classes (Healthy, CSSVD, and Anthracnose) were chosen, providing a final number of 7,752 images. Three fine-tuned CNN architecture (DenseNet121, ResNet50, and EfficientNetB0) were trained with transfer learning. All models used uniform hyper-parameters and five-fold cross-validation was used to evaluate their robustness.

Of the three models, DenseNet-121 received the highest performance in terms of the test accuracy of 90% and the most balanced precision, recall, and F1-score in each of the classes. It was also the most memorable to CSSVD and therefore it can be utilized in places where a wrong diagnosis may result in significant losses of production. The stability of DenseNet121 was also supported by cross-validation in which Fold 3 presented the best result. A lightweight setup has been used with its deployment to the model, including Kaggle Notebooks for backend, ngrok for API tunnelling, and Vercel for frontend hosting. The resulting web application enables the users to upload or take images of cocoa leaves, get predictions of the diseases, and get the real time recommendations.

5.2 Recommendations

In light of the findings of this report, the following suggestions are proposed:

- a. Future research should involve the expansion of the dataset by adding additional classes of diseases which also affect the cocoa leaf such as Black Pod and

Witches' Broom and gathering images in different environmental conditions to enhance the model generalization.

- b. The web application delivered should be put through field test with actual farmers and extension personnel to be able to test ease of use and feedback on improvements to be done.
- c. Online coverage is inadequate in most areas where farmers are located, and therefore a native mobile application should be designed to allow it to work offline.
- d. To keep the performance stable and even enhance it with the course of time, continuous collecting of data and periodical retraining of the model should be offered.
- e. Collaboration should be transacted with agricultural agencies, research institutes and local cooperatives to aid real-world uptake and implementation.

5.3 Conclusions

This study has adequately shown how CNNs can be used to carry out cocoa disease diagnosis with leaf images automatically and more effectively. The choice of DenseNet121 allowed the system to obtain high values of classification accuracy and equalize performance results on the most important categories of disease. The study also offers a gap between efficient AI models and real-life utilization through the deployment of the model in the form of a web app available on mobiles and desktops.

Overall, the proposed project is cost-effective, adaptable, and easy to implement, and could help cocoa farmers detect and address diseases in time, avoiding the need to involve experts and the losses because of the diseases. This is an illustration of the possibility to implement artificial intelligence in agriculture to enhance its productivity and sustainability.

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